Mechanism Design with Costly Inspection

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This paper studies how to combine screening menus and inspection in mechanism design. A Principal procures a good from an Agent whose cost is his private information. The Principal has two instruments: screening menus — i.e., quantities and transfers — and (ex-ante) inspection. Inspection is costly but reveals the Agent's cost. The combination of inspection and screening menus mitigates inefficiencies: the optimal mechanism procures an efficient quantity from all Agents whose cost of production is sufficiently low, regardless of whether inspection has taken place. However, quantity distortions still necessarily occur in optimal regulation; the quantity procured from Agents with higher production costs is inefficiently low. In contrast to settings without inspection, incentive compatibility constraints do not bind locally. Nonetheless, the paper characterizes which incentive constraints bind.
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1. Introduction

Contemporary approaches to regulation operate under the assumption that regulators have less information about important data than the entity targeted by the regulation. For instance, a government lacks knowledge about a defense manufacturer's production costs, or a municipality is uninformed about the cost of constructing new infrastructure. The literature on regulation and procurement shows how to design transfer schemes in order to alleviate the problems caused by such asymmetric information.

In practice, however, regulators have the capabilities to acquire direct knowledge of the unknown variable. For example, the Defense Contract Audit Agency (DCAA), an agency under the United States Department of Defense, conducts inspections that "are generally completed before contract award where DCAA evaluates [...] how much it will cost the contractor to provide goods or services to the government."[1] The U.S. General Service Administration has its own Office of Audits whose responsibilities include conducting audits in procurement cases and construction projects. Its "[a]udits of [construction projects] take place before a contract is awarded" and "include the evaluation of submitted cost or pricing data[...]."[2]

In this paper, we take a mechanism design approach to regulation and allow the regulator to use transfers, inspection and quantity menus. We examine the trade-offs between using either instrument and characterize their optimal use in regulation.

In our model, an Agent produces a good that the Principal values. The Agent has private information about his cost of production, a continuous variable. The Principal aims to procure a cost-dependent quantity from the Agent. To elicit the Agent's cost, the Principal designs a transfer and quantity scheme, and she can also learn the Agent's cost through costly inspection.

The Principal offers a mechanism to the Agent. For each reported cost, a (direct) mechanism specifies a probability that the Agent is inspected, as well as contingent quantities and transfers. More precisely, the mechanism specifies a quantity the Agent produces when not inspected and a transfer he receives as a function of the reported cost. When inspected, the mechanism specifies a quantity and a transfer conditional on the reported and the true cost, which has been observed through inspection. Crucially, the Agent is free to reject the mechanism ex–post and collect his outside option instead. This means the Principal cannot force the Agent to produce without reimbursing his cost of production. We place no further restrictions on the mechanism, including the magnitude of the transfers, and allow for stochastic inspection.

In the first part of our analysis, we solve for an optimal mechanism under the restriction that the Principal has to inspect each report with a minimal probability. An optimal mechanism provides incentives to the Agent only through inspection and bonus payments when the Agent has reported his cost truthfully. In case the Agent is not inspected, the Principal reimburses him his reported cost of production. Payments after inspection are strictly higher than the Agent's cost when he has reported his true type.

[1] See DCAA (2023, p. 4).
[2] U.S. General Services Administration (2012, p. 12).

This payment scheme is optimal because it minimizes the rents an Agent can extract through misreporting. In particular, it enables the Principal to reduce information rents relative to the case in which inspection is infeasible. When inspection is infeasible, the Principal must offer transfers strictly above the Agent's true cost to implement a non-trivial allocation. Such transfers allow the Agent to extract additional rents by misreporting and receiving payments above their reported cost. By contrast, under the optimal mechanism with inspection, the transfer equals the reported cost, thereby reducing information rents. We illustrate this with Example 1, presented in Section 4.1.

Providing incentives solely through bonus payments after inspection and truthful reports implies that incentive constraints do not bind locally, but only globally. If the Agent reports a cost slightly above his true cost, he forgoes the bonus but secures a payment proportional to the difference between the reported and true costs when not inspected. Consequently, the Agent is indifferent between truthful reporting and overstating his cost only when the misreport is sufficiently large. Figure 1 shows the binding incentive constraints in the optimal mechanism: the Agent is indifferent between reporting his true cost θ and reporting $\hat{\theta}(\theta)$, but strictly prefers either to any alternative report.[3]

Quantities and inspection probabilities in the optimal mechanism follow a two-threshold policy. Specifically, there exist two cost thresholds: a "distortion threshold" θ_1, which lies strictly above the lowest cost type, and an "inspection threshold" θ_2, with $\theta_1 < \theta_2$. Figure 2 illustrates this policy. The first panel depicts the inspection probability $x(\theta)$ for a report θ, while the second panel shows the quantity $q^N(\theta)$ the Agent produces when not inspected after reporting θ.

The Agent produces the first–best quantity even when not inspected if his cost is below the distortion threshold. While reminiscent of the familiar "no–distortion–at–the–top"-result in screening problems, this is a stronger property: an interval of types produces the first-best quantity regardless of whether they are inspected or not. Hence, there is no downward distortion of quantities for an interval of low-cost types. Because incentive constraints only bind upwards and local incentive constraints are slack, no type is indifferent between reporting truthfully and reporting a cost in a region of low-cost types. These are the types θ below the threshold θ_1 in Figure 1. Distorting quantities for this region of types does not affect information rents, and is thus strictly sub-optimal. The quantity procured from types above the distortion threshold is strictly below the first–best benchmark.

The Agent is inspected with probability above the lower bound only if he reports a cost above the inspection threshold θ_2. Low-cost types are inspected with the minimal probability since inspecting them with a higher probability does not reduce information rents. Intermediate cost types—those between the distortion and the inspection threshold—are inspected with the minimal probability as well: it is more costly to reduce information rents through increasing the inspection probability than through quantity distortions as long as these distortions are small enough. Furthermore, no type is inspected with prob-

[3]The dotted line shows the binding incentive constraints when inspection is not feasible. In that case, local incentive constraints bind.

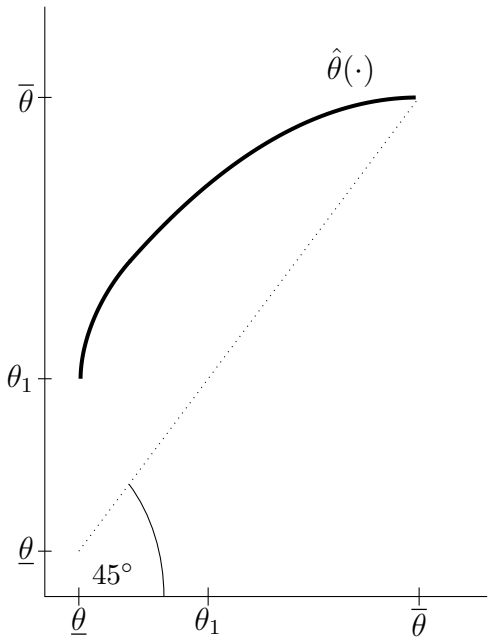


Figure 1: Binding incentive constraints in the constrained optimal mechanism of Example 1. Type θ on the vertical axis is indifferent between reporting his type truthfully and reporting $\hat{\theta}(\theta)$. The dotted line shows the binding incentive constraints in the optimal mechanism when inspection is infeasible and, consequently, local incentive constraints bind.

ability one as this is too costly for the Principal. This shows that restricting attention to deterministic inspection policies would result in a loss of optimality.[4]

In designing a mechanism, the Principal has two instruments to reduce information rents of the Agent: inspecting the Agent with higher probability or decreasing the quantity when not inspected. We show that, unless inspection costs are prohibitively large, the Principal uses both instruments: high cost types are inspected with positive probability and produce an inefficiently low quantity.

In the second part of our analysis, we drop the restriction that the Principal has to inspect each report with a minimal probability. As is common in models with inspection and unbounded transfers, no optimal mechanism exists. However, we show that for all ε-optimal mechanisms, the limits (as ε goes to 0) of the quantity allocation and inspection policy are unique. Moreover, these limits coincide with the limit quantities and inspection probabilities of the optimal mechanism derived in the first part of our analysis, as the lower bound on the inspection probability vanishes.

Our results highlight the efficiency implications of combining screening menus with costly inspection. When inspection takes place before production, and quantities and

[4]In Section 4.4, we show that reporting higher costs need not trigger inspection with a higher probability: there exist parameter values such that the inspection probability is not monotone in the Agent's reported cost; see Example 2. We also provide sufficient conditions for the inspection probability to be monotonically increasing in the reported cost.

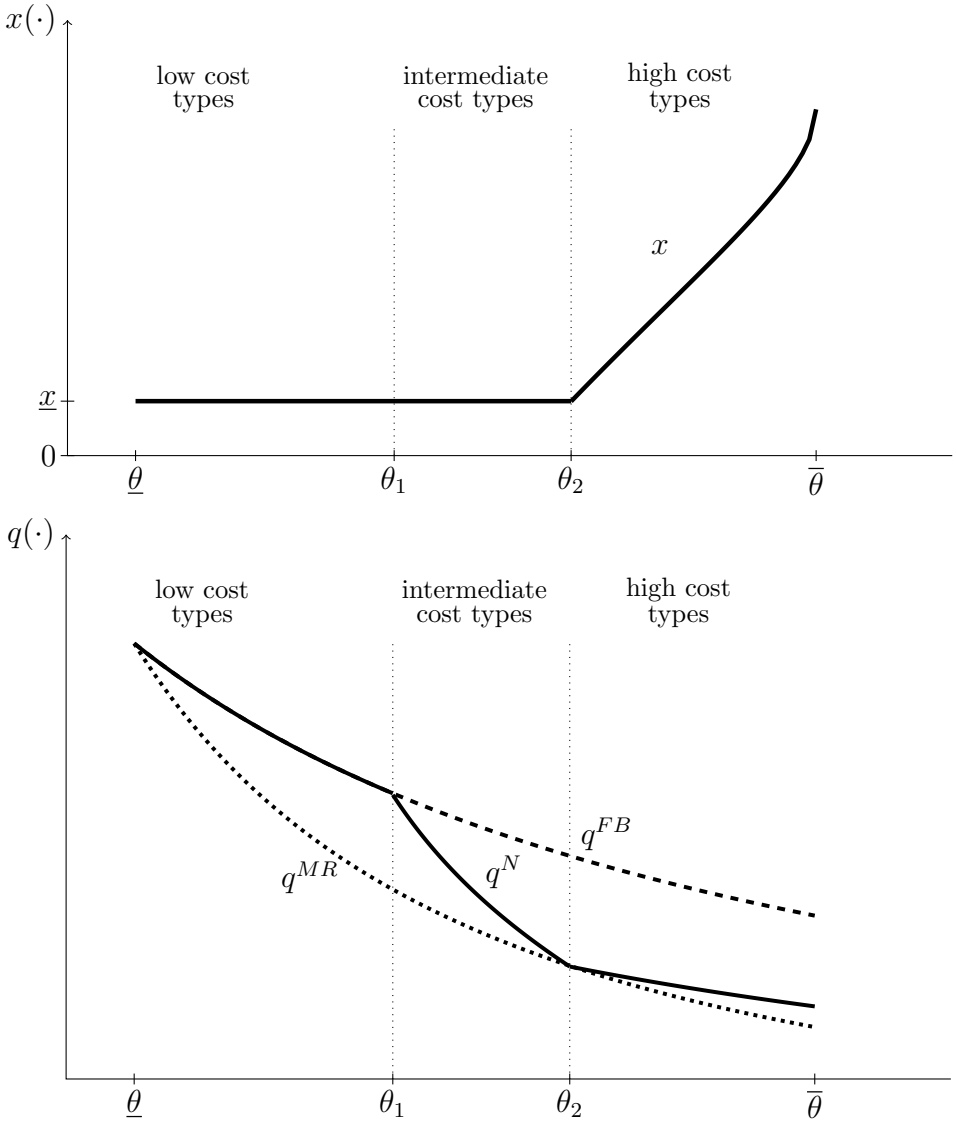


Figure 2: The mechanism of Theorem 1 for the parameter values of Example 1. $q^N(\cdot)$ is the quantity without inspection. $q^{MR}(\cdot)$ is the optimal quantity in a setting where inspection is not feasible (see Mussa and Rosen (1978)).

payments can be made contingent on its outcome, inefficiencies are mitigated: the optimal mechanism procures an efficient quantity from all agents whose cost of production is sufficiently low. However, quantity distortions still necessarily occur in optimal regulation; the quantity procured from agents with higher production costs is inefficiently low.

From a methodological point of view, we combine the literatures on costly state verification and monopolistic screening. The literature on costly state verification has focused on the trade-off between costly information rents and costly inspection. The literature on monopolistic screening has emphasized the role of transfers and quantity distortions in providing incentives. Our model combines both aspects. It allows us to study the trade-off between quantity distortions and inspection costs in providing incentives to the Agent.

As mentioned, a major challenge is that, in our problem, incentive constraints do not bind locally, but only globally. Consequently, we cannot employ standard techniques based on the envelope theorem or the first-order approach. Nonetheless, we are able to characterize *which* incentive constraints bind in an optimal mechanism. We characterize the binding incentive constraints as the unique solution to a system of differential equations. This characterization is key to determining the optimal quantities, transfers and inspection probabilities.

The rest of the paper is organized as follows. The next section discusses our contribution to the literature. Section 3 introduces the model. Section 4 analyses the model for a fixed positive lower bound on the inspection probability. In Section 5, we analyse the model without imposing a lower bound on the inspection probability. Proofs are relegated to the Appendix. Section 6 concludes.

2. Related Literature

Our paper combines monopolistic screening (Mussa and Rosen (1978)) or procurement under incomplete information (Baron and Myerson (1982)) with costly state verification (Townsend (1979), Gale and Hellwig (1985)). The papers closest to ours are Baron and Besanko (1984) and Palonen and Pekkarinen (2022).[5] The relevant differences in the modeling assumptions between Baron and Besanko (1984) and our paper are the following three. First, Baron and Besanko (1984) impose a bound on transfers. Second, they set the Agent's payoff after inspection equal to his outside option; in particular, the Agent cannot be rewarded for reporting truthfully. Lastly, inspection in their model is imperfect whereas it is perfect in our paper. Moreover, Baron and Besanko (1984, see pp. 464) solve for the optimal mechanism under the assumption that local incentive constraints are sufficient for global incentive compatibility. While they "have been unable to extend the continuous-type analysis to deal with this problem" (p. 465), they provide a sufficient condition for local incentive compatibility to imply global incentive compatibility, as well as a numerical example (on pp. 465) that suggests their results

[5]Khalil (1997) studies a similar problem, but assumes the Principal cannot commit to inspect the Agent.

no longer hold if global incentive compatibility is taken into account. In contrast, we show that local incentive constraints never bind, and deal explicitly with global incentive compatibility.

Palonen and Pekkarinen (2022) study a model in which the Agent can exert costly effort to reduce the probability of being inspected. In order to focus on this avoidance activity, they assume that the allocation and transfer in case of inspection are exogenous. By contrast, in our paper, the Principal designs a mechanism and commits to the allocation and transfer after inspection. The allocation and transfer Palonen and Pekkarinen (2022) impose are with loss of optimality for the Principal in our setting, and imply that local incentive constraints bind. We compare our results to those in Baron and Besanko (1984) and in Palonen and Pekkarinen (2022) after presenting the optimal mechanism.

Border and Sobel (1987), Mookherjee and Png (1989), and Chander and Wilde (1998) study wealth extraction with audits. In their models, there is no allocation or production of a good other than monetary transfers. Border and Sobel (1987) and Chander and Wilde (1998) impose an upper bound on transfers from the Principal to the Agent; we do not impose an upper bound on transfer. Mookherjee and Png (1989) assume that the Agent is risk-averse, which ensures the existence of an optimal mechanism. Border and Sobel (1987), Mookherjee and Png (1989), and Chander and Wilde (1998) find that, for a truthful report, the Agent's payoff is higher after being inspected than when not inspected — a result that obtains in our setting as well. Mookherjee and Png (1989) find that no report is inspected with probability 1, as we do, and in contrast to Chander and Wilde (1998). All three papers do not explicitly characterize the optimal mechanism. With the exception of the threshold θ_1, we fully characterize the optimal mechanism. Melumad and Mookherjee (1989) study taxation with audits and the provision of a public good. In their model, all agents consume the same quantity of the public good whereas in our model the allocation is type-dependent. Moreover, they rule out transfers to the Agent after truthful reports.

Dana, Larsen, and Moshary (2024) study a mechanism design problem in which the Agent has a uniform cost of misreporting his true type. This leads to incentive constraints that do not bind locally. In our model, incentive constraints do not bind locally for a different reason: transfers equal the cost of production when the Agent is not inspected. Optimal mechanisms in Dana, Larsen, and Moshary (2024) have the "no-distortion-at-the-top"-property of Theorem 1.

Alaei et al. (2024) study an auction model in which the auctioneer can inspect the bidder's valuation ex–post.[6] There are three differences to our model: first, Alaei et al. (2024) assume inspection is costless to the Principal and hence all types are inspected with probability 1; second, in their paper, the allocation of the good depends on the reported type, but not on the true type; third, their auctioneer has no valuation for the good.[7]

[6] See also the literature on auctions with contingent payments as surveyed in Skrzypacz (2013).

[7] Alaei et al. (2024) deal with global incentive constraints in a different way than we do. They convert their problem of finding the optimal quantity into a maximization problem with the binding incentive constraints as the choice variable. This inversion step (their Proposition 3) cannot be applied in

3. Model

There is a Principal ("she") and an Agent ("he"). The Agent produces a good that the Principal aims to procure. The Agent's per unit cost of production, his type, $\theta \in [\underline{\theta}, \overline{\theta}], 0 < \underline{\theta} < \overline{\theta}$, is private information. The Principal believes the Agent's type is distributed according to the prior F. We assume the distribution F admits a density f that is continuously differentiable and bounded away from 0. The Principal aims to induce a type-contingent quantity allocation $q : [\underline{\theta}, \overline{\theta}] \to \mathbb{R}_+$.

The Principal has the ability to inspect the Agent's type. When the Principal inspects the Agent, she learns his type perfectly. Inspection costs $\kappa > 0$ to the Principal. Moreover, the Principal can pay a transfer to the Agent.

The utility of the Agent from an allocation q and a transfer t given his type θ is $-\theta q + t$. The utility of the Principal is $V(q) - t - \kappa \mathbb{1}_{\text{inspection}}$. We make the following standard assumption about the Principal's preferences.

Assumption 1. *$V : \mathbb{R}_+ \to \mathbb{R}_+$ is twice continuously differentiable with $V' > 0$, $V'' < 0$, and satisfies the Inada conditions $V'(q) \to_{q \searrow 0} \infty$ and $V'(q) \to_{q \to \infty} 0$.*

The Principal offers a mechanism $\mathbb{M}$ to the Agent. It is without loss to focus on direct mechanisms.[8] A direct mechanism is a tuple $\mathbb{M} = \big(x(\cdot), q^I(\cdot,\cdot), t^I(\cdot,\cdot), q^N(\cdot), t^N(\cdot)\big)$. Here, $x(\hat{\theta})$ denotes the probability that the Agent is inspected when he reports type $\hat{\theta}$. When the Agent is inspected, the Principal learns his type θ, and the Agent produces $q^I(\hat{\theta}, \theta)$ and receives the transfer $t^I(\hat{\theta}, \theta)$. Note that quantity and transfer after inspection depend both on the Agent's report $\hat{\theta}$ and his true type θ, which the Principal has learned through inspection. When the Agent is not inspected, he produces the quantity $q^N(\hat{\theta})$ and gets paid the transfer $t^N(\hat{\theta})$, conditional only on his report.

We assume that the Agent can reject the mechanism ex–post:[9] after the Agent observes if he has been inspected and observes the quantity he needs to produce as well as the transfer he receives, the Agent can walk away and secure a payoff of 0. When the Agent walks away from the mechanism, no production takes place, and the Principal does not pay the transfer. Standard arguments show that it is without loss of generality to focus on mechanisms that induce the Agent to not walk away. This is also without loss of optimality, as the Principal can improve on any mechanism that induces the Agent to reject on path. Throughout, we assume that the Agent does not reject the mechanism when indifferent. A direct mechanism that the Agent does not reject ex-post

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our setting: a fixed set of binding incentive constraints $\hat{\theta}(\cdot)$ does not uniquely pin down both the quantity without inspection $q^N(\cdot)$ and the inspection probabilities $x(\cdot)$.

[8] See Appendix A in Mookherjee and Png (1990).

[9] Assuming the Agent can reject the mechanism ex-post puts a lower bound on the payoff the Agent receives. Without a lower bound on the Agent's payoff after inspection, the Principal is able to implement the first-best quantity at a cost arbitrarily close to 0 by inspecting the Agent with a vanishingly small probability and driving the Agent's payoff to $-\infty$ if the Agent has not reported his true type (Becker (1968)). The assumption that the Agent can reject the mechanism ex-post is stronger than putting a lower bound on transfers. Indeed, the Principal could make the Agent's payoff arbitrarily small even with bounded transfers by requiring an arbitrarily large quantity.

must satisfy, for every type θ and report $\hat{\theta}$, the following *ex-post individual rationality constraints*,

$$-q^I(\hat{\theta},\theta)\theta + t^I(\hat{\theta},\theta) \geq 0; \tag{IR-I}$$
$$-q^N(\theta)\theta + t^N(\theta) \geq 0. \tag{IR-N}$$

The Principal's problem is to choose a mechanism $\mathbb{M}$ that induces truth-telling and satisfies the ex-post individual rationality constraints. Formally, the Principal's problem is:

$$\begin{aligned}
\sup_{\mathbb{M}} \quad & \int_{\underline{\theta}}^{\overline{\theta}} \left\{ x(\theta)\Big(V(q^I(\theta,\theta)) - t^I(\theta,\theta) - \kappa\Big) + (1-x(\theta))\Big(V(q^N(\theta)) - t^N(\theta)\Big)\right\} \mathrm{d}F(\theta) \\
& \text{subject to, for all } \theta, \hat{\theta} \\
& x(\theta)\left(-q^I(\theta,\theta)\theta + t^I(\theta,\theta)\right) + (1-x(\theta))\left(-q^N(\theta)\theta + t^N(\theta)\right) \\
& \geq x(\hat{\theta})\left(-q^I(\hat{\theta},\theta)\theta + t^I(\hat{\theta},\theta)\right) + (1-x(\hat{\theta}))\left(-q^N(\hat{\theta})\theta + t^N(\hat{\theta})\right); \qquad \text{(IC)} \\
& -q^N(\theta)\theta + t^N(\theta) \geq 0; \\
& -q^I(\hat{\theta},\theta)\theta + t^I(\hat{\theta},\theta) \geq 0; \\
& \underline{x} \leq x(\theta) \leq 1.
\end{aligned}$$

The first inequality constraint is the incentive compatibility constraint for type θ. It requires that type θ prefers reporting his true type θ to reporting any other type $\hat{\theta}$. The next two inequalities are the ex-post individual rationality constraints: the Agent prefers honoring the mechanism to walking away, both after inspection and without inspection. The last inequality says that the inspection probability is between a fixed lower bound $\underline{x}$ and 1.

Denote the problem by $\mathcal{P}_{\underline{x}}$ and its value by $W_{\underline{x}}$; for $\underline{x} = 0$, call the problem "unconstrained", for $\underline{x} > 0$ call the problem "constrained". We first analyze the constrained problem in Section 4 before analyzing the unconstrained problem in Section 5.

Under Assumption 1, for every type θ, the first-best quantity

$$q^{FB}(\theta) = \arg\max_q \; V(q) - q\theta$$

exists and is unique. We assume that there is no interval (θ', θ'') such that $q^{FB}(\theta) = 1/(c_1\theta - c_2)$ for all $\theta \in (\theta', \theta'')$ for positive constants $c_1 > 0, c_2 \in [\underline{\theta}, \overline{\theta}]$.[10]

4. Constrained Problem

In this section, we analyze the Principal's problem for a fixed positive lower bound on the inspection probability, $\underline{x} > 0$. Section 4.1 derives the optimal transfer schemes

[10] The assumption holds if there is no interval (q_1, q_2) such that $V(q) = c_1 \ln(q) + c_2 q + c_3$ for constants $c_1, c_2 > 0, c_3$ and all $q \in (q_1, q_2)$. It is generically satisfied. We return to the significance of this assumption in Section 4.2.

and shows that incentives are provided through payments after inspection and truthful reports. Section 4.2 derives the quantities and inspection probabilities in the constrained optimal mechanism. In Section 4.3, we show how we deal with non-local incentive constraints.

4.1. Providing incentives

We start the analysis with an example that illustrates how the use of transfers contingent on the outcome of inspection allows the Principal to reduce information rents compared to a setting where inspection is not feasible.

Example 1. *Suppose that* $V(q) = \ln(q), [\underline{\theta}, \overline{\theta}] = [1, 2]$, *and* $f(\theta) = 1/(\theta \ln(2))$. *Suppose for now that the Principal cannot inspect the Agent's report. The optimal quantity* $q^{MR}(\theta)$ *for a type* θ *is given by (see Mussa and Rosen (1978))*

$$V'(q^{MR}(\theta)) = \theta + F(\theta)/f(\theta) \iff q^{MR}(\theta) = 1/(\theta(1 + \ln(\theta)).$$

Together with the transfers $t^{MR}(\theta) = 1/(1 + \ln(\theta)) + \ln(1 + \ln(2)) - \ln(1 + \ln(\theta))$, *the mechanism* $(q^{MR}(\cdot), t^{MR}(\cdot))$ *is incentive compatible and satisfies* (IR-N). *The total transfer the Principal pays in excess of the production cost is approximately* 0.211.

Now suppose that the Principal can inspect the Agent and make transfers contingent on the outcome of inspection. For the sake of illustration, suppose each type is inspected with the minimal probability, $x(\theta) = \underline{x}$. *The Principal can then set the transfer without inspection equal to the cost of production,* $t^N(\theta) = \theta q^{MR}(\theta)$, *so that* (IR-N) *binds. The transfer after inspection when the Agent has not reported his true type is set to make* (IR-I) *bind,* $t^I(\hat{\theta}, \theta) = \theta q^{MR}(\hat{\theta})$ *for each* $\hat{\theta} \neq \theta$. *The transfer after inspection and truthful reporting is set to make the incentive constraint bind:*

$$\underline{x}\left(t^I(\theta, \theta) - \theta q^{MR}(\theta)\right) = \sup_{\hat{\theta}} (1 - \underline{x}) q^{MR}(\hat{\theta})(\hat{\theta} - \theta).$$

The LHS is type θ*'s payoff from reporting truthfully and the RHS is the highest payoff the Agent can achieve by reporting a type* $\hat{\theta} \neq \theta$. *The total expected transfer the Principal pays in excess of production costs is* $(1-\underline{x})1/(4(1+\ln(2))) \approx (1-\underline{x})0.148$. *This is lower than the transfers required without inspection for any* $\underline{x} > 0$. *The Principal is better off inspecting each report with probability* $\underline{x}$ *than without having the possibility to inspect if* $(1 - \underline{x})0.148 + \underline{x}\kappa < 0.211$. *No matter how high the inspection cost* κ *is, there exists a* $\underline{x} > 0$ *such that the Principal is better off when she has the opportunity to inspect the Agent's report.*[11]

The Principal can further improve upon her payoff by (i) implementing different quantities with and without inspection, (ii) implementing quantities different from $q^{MR}(\cdot)$, *and (iii) making the inspection probability dependent on the reported type.*

We start the analysis by determining the transfers in an optimal constrained mechanism.

[11] See Proposition 4 for a formal statement.

Proposition 1. *1. It is without loss of generality to consider mechanisms that punish misreports as harshly as possible: for all $\theta \neq \hat{\theta}$, (IR-I) binds, i.e.,*

$$t^I(\hat{\theta}, \theta) = \theta q^I(\hat{\theta}, \theta) \text{ for all } \hat{\theta} \neq \theta.$$

2. For every incentive compatible mechanism that satisfies the constraints (IR-I) and (IR-N) there exists an incentive compatible mechanism such that the transfer without inspection equals the cost of production, i.e., (IR-N) binds,

$$t^N(\theta) = \theta q^N(\theta),$$

and both mechanisms have the same quantity allocation and inspection probability. Moreover, both mechanisms yield the same payoff to the Principal.

3. For any bounded quantity allocation $(q^I(\cdot,\cdot), q^N(\cdot))$ and inspection policy $x(\cdot)$, there exist transfers $(t^I(\cdot,\cdot), t^N(\cdot))$ such that the mechanism is incentive compatible and satisfies the (IR-I) and (IR-N) constraints. Moreover, it is without loss of optimality to focus on transfers such that for all types θ, (IR-N) binds and incentives are provided after inspection, i.e.,

$$x(\theta)\left(t^I(\theta,\theta) - \theta q^I(\theta,\theta)\right) = \sup_{\hat{\theta}}(1 - x(\hat{\theta}))q^N(\hat{\theta})(\hat{\theta} - \theta). \tag{1}$$

Proof in Appendix A.1.

Proposition 1 pins down the optimal transfer schemes for any quantity allocation and inspection policy. Recall that the incentive constraint for type θ reads

$$\begin{aligned} &x(\theta)\left(-q^I(\theta,\theta)\theta + t^I(\theta,\theta)\right) + (1 - x(\theta))\left(-q^N(\theta)\theta + t^N(\theta)\right) \\ \geq\ &x(\hat{\theta})\left(-q^I(\hat{\theta},\theta)\theta + t^I(\hat{\theta},\theta)\right) + (1 - x(\hat{\theta}))\left(-q^N(\hat{\theta})\theta + t^N(\hat{\theta})\right) \text{ for all } \hat{\theta}. \end{aligned}$$

Observe that the transfer after inspection, $t^I(\hat{\theta}, \theta)$, for $\hat{\theta} \neq \theta$ does not enter the incentive constraints for any type $\theta' \neq \theta$. Therefore, reducing $t^I(\hat{\theta}, \theta)$ relaxes the incentive constraints. Moreover, $t^I(\hat{\theta}, \theta)$ and $q^I(\hat{\theta}, \theta)$ do not affect the Principal's payoff. Therefore, it is without loss to restrict attention to mechanisms that make (IR-I) bind after a misreport: for every θ,

$$-q^I(\hat{\theta}, \theta)\theta + t^I(\hat{\theta}, \theta) = 0 \ \forall \hat{\theta} \neq \theta.$$

This is intuitive: it is optimal to punish the Agent for misreporting his type as harshly as possible. In the remainder of the paper, we abuse notation and write $t^I(\theta)$ and $q^I(\theta)$ instead of $t^I(\theta, \theta)$ and $q^I(\theta, \theta)$, respectively.

Denote by $\pi(\theta)$ the rent of type θ, i.e., the highest payoff type θ can obtain by reporting any type $\hat{\theta}$. The first part of Proposition 1 implies that the Agent receives a payoff above 0 when he misreports his type only in case he is not inspected. When he is not inspected, his payoff is the transfer he receives minus his cost of production, $t^N(\hat{\theta}) - q^N(\hat{\theta})\theta$. Consequently, the information rent of type θ is

$$\pi(\theta) = \sup_{\hat{\theta}}(1 - x(\hat{\theta}))(-q^N(\hat{\theta})\theta + t^N(\hat{\theta})).$$

Each type θ is inspected with positive probability. Because the Agent is risk–neutral with respect to transfers, his incentive to report his true type depends solely on the expected transfer (less the cost of production), $x(\theta)t^I(\theta) + (1 - x(\theta))t^N(\theta)$. Therefore, his incentives are preserved if a decrease in the transfer without inspection, $t^N(\theta)$, is compensated by an increase in the transfer after inspection, $t^I(\theta)$, such that the expected transfer remains constant. Because the Principal is risk-neutral with respect to transfers as well, this change leaves her payoff unaffected. Reducing the transfer without inspection that is paid to type θ, however, reduces the information rent that has to be paid to types $\theta' \neq \theta$. Consequently, for any incentive-compatible mechanism, there is a payoff-equivalent mechanism, implementing the same quantity allocation, such that the Agent gets paid his reported cost of production when not inspected. This shows the second part of the Proposition.

The second part of Proposition 1 also implies that restricting attention to mechanisms that satisfy $t^N(\theta) = \theta q^N(\theta)$ does not rule out any multiplicity in the quantity allocation or inspection probability in a constrained optimal mechanism.[12] We henceforth focus on mechanisms such that (IR-N) binds for all types. In such a mechanism, the Agent's payoff is strictly higher when inspected than when not inspected. Similar results appear already in the tax-audit literature; see Theorem 1, equation (5.2a) in Border and Sobel (1987), Proposition 1, part c), of Mookherjee and Png (1989), and equation (11) in Chander and Wilde (1998). The similarity to the tax-audit literature should come as no surprise. Proposition 1 does not depend on the quantity allocation. Hence, results similar to settings without quantities, as is the case in the tax-audit literature, are to be expected.

The third part of Proposition 1 says that any allocation with bounded quantities $(q^I(\cdot), q^N(\cdot))$ and inspection policy $x(\cdot)$ can be made incentive-compatible. The restriction to bounded quantity allocations is, of course, without loss of optimality. The reason that any allocation can be implemented is that transfers after inspection allow the Principal to provide targeted incentives: the transfer after inspection and truth-telling, $t^I(\theta)$, cannot be appropriated by any other type $\theta' \neq \theta$. Consequently, changing transfers after inspection $t^I(\theta)$ does not affect incentives for types $\theta' \neq \theta$, and for each allocation can be set to induce type θ to report truthfully.

The third part also says that it is optimal for the Principal to provide all incentives through inspection and transfers after truthful reports. Since no type other than θ can get the transfer $t^I(\theta)$, increasing it does not jeopardize incentives for other types θ'. Hence, it is optimal to provide all rents after inspection and truth-telling by reducing the transfers without inspection as much as possible, i.e., until (IR-N) binds.

Using that $-q^N(\theta)\theta + t^N(\theta) = 0$, it is straightforward to pin down the transfer after inspection. Because slack incentive constraints for some types θ cannot be optimal, the transfer after inspection must satisfy

$$x(\theta)\left(-q^{FB}(\theta)\theta + t^I(\theta)\right) = \sup_{\hat{\theta}} (1 - x(\hat{\theta}))q^N(\hat{\theta})(\hat{\theta} - \theta).$$

Here, we used $-q^N(\theta)\theta + t^N(\theta) = 0$ on the left–hand side of (IC) for type θ as well as

[12] Proposition 2 characterizes all transfers that feature in a constrained optimal mechanism.

on the right-hand side for type $\hat{\theta}$. To implement a monotone allocation $q^N(\cdot)$ without inspection, transfers must equal $t^N(\theta) = \theta q^N(\theta) + \int_\theta^{\overline{\theta}} q^N(s)\,\mathrm{d}s$. Consequently, the Agent's information rent is

$$\pi^{MR}(\theta) = \sup_{\hat{\theta}} q^N(\hat{\theta})(\hat{\theta} - \theta) + \int_{\hat{\theta}}^{\overline{\theta}} q^N(s)\,\mathrm{d}s,$$

which is strictly higher than the information rents the Principal needs to deliver to the Agent under the optimal transfer scheme when inspection is feasible. We remark that information rents are lower when inspection is feasible for any inspection policy such that each report is inspected with positive probability.

Proposition 1 holds for any inspection policy $x(\cdot)$ such that each type is inspected with positive probability, $x(\theta) > 0$ for each type θ. Shifting incentives to transfers after inspection is optimal for each inspection policy. Transfers without inspection are independent of the inspection policy. Transfers after inspection, $t^I(\cdot)$, depend on the inspection policy $x(\cdot)$ both through the right-hand and the left-hand side of equation (1).

Essentially, for any quantities $(q^I(\cdot), q^N(\cdot))$ and inspection policy $x(\cdot)$, the optimal transfer scheme subjects the Agent to a lottery. When the Agent reports truthfully, he receives a payoff of 0 when not inspected, but a bonus, $t^I(\theta) - \theta q^I(\theta)$, above his production cost when inspected. When the Agent misreports his type, he receives a payoff depending on the difference between the true and the reported cost when not inspected, but a payoff of 0 when inspected. Such a transfer scheme implies that incentive constraints do not bind locally, but only globally. When the Agent reports truthfully, he gets a bonus when inspected. By reporting a cost $\hat{\theta} = \theta + \delta$ above his type θ, the Agent gives up the chance to receive this bonus. Instead, he engages in a gamble to receive $\delta q^N(\hat{\theta})$ when not inspected and 0 otherwise. Engaging in this gamble is worthwhile only if the difference between the true and the reported cost δ is large enough.

More formally, (1) implies that the payoff of type θ is $\pi(\theta) = \sup_{\hat{\theta}}(1 - x(\hat{\theta}))q^N(\hat{\theta})(\hat{\theta} - \theta)$ when reporting truthfully. This payoff is zero if and only if for all types $\theta' > \theta$, $(1 - x(\theta'))q^N(\theta') = 0$. The quantity without inspection $q^N(\theta)$ is bounded from above by the first–best quantity.[13] Thus, for every type θ that receives a positive payoff there exists a $\delta > 0$ such that $(1 - x(\hat{\theta}))q^N(\hat{\theta})(\hat{\theta} - \theta) < \pi(\theta)$ for all $\hat{\theta}$ that are within δ of θ. Hence, reporting a type $\hat{\theta}$ within δ of θ gives type θ a payoff strictly less than what he gets from reporting truthfully.

A second implication is that incentive constraints only bind upwards: reporting a type $\hat{\theta} < \theta$ yields a negative payoff. Hence, downward incentive constraints are slack for each type that receives a positive payoff.

Having pinned down the optimal transfer schemes for any bounded allocation $(q^I(\cdot), q^N(\cdot))$

[13] We state this formally in Theorem 1.

and inspection policy $x(\cdot)$, we have reduced the constrained problem to

$$\max_{q^I(\cdot),q^N(\cdot),\underline{x}\leq x(\cdot)\leq 1} \int \Big\{ x(\theta)\left(V(q^I(\theta)) - q^I(\theta)\theta - \kappa\right) + (1-x(\theta))\left(V(q^N(\theta)) - q^N(\theta)\theta\right) - \sup_{\hat{\theta}}(1-x(\hat{\theta}))q^N(\hat{\theta})(\hat{\theta}-\theta)\Big\}\,\mathrm{d}F(\theta) \qquad \text{(reduced problem)}$$

The term in the first line is the social welfare from an allocation (x, q^N, q^I). The term on the second line is the information rent.

In designing the mechanism, the Principal faces a trade-off between quantity distortions, higher information rents and higher inspection costs. Both higher inspection probabilities and lower quantities without inspection reduce information rents weakly. However, they come at the expense of higher inspection costs and lower utility from the quantity allocation, respectively.

4.2. Optimal quantity and inspection probability

In this section, we describe the quantities and the inspection probability in a constrained optimal mechanism for a fixed lower bound $\underline{x} > 0$ on the inspection probability.

For a fixed type $\theta_1 \in (\underline{\theta}, \overline{\theta})$ let $(q_1, \hat{\theta}_1)$ be the solution to

$$-q'(\hat{\theta})(\hat{\theta}-\theta) = q(\hat{\theta}), \tag{2}$$

$$(\hat{\theta}(\theta)-\theta)f(\theta) = \hat{\theta}'(\theta)\left(V'(q(\hat{\theta}(\theta))) - \hat{\theta}(\theta)\right)f(\hat{\theta}(\theta)), \tag{3}$$

with the boundary conditions $q_1(\theta_1) = q^{FB}(\theta_1), \hat{\theta}_1(\underline{\theta}) = \theta_1$.

With $q_1(\theta)$ as defined above, denote

$$\theta_2 = \min\{\overline{\theta}, \inf\{\theta | q_1(\theta) < q_2(\theta)\}\}, \tag{4}$$

where $q_2(\theta)$ is the solution to (5).

Denote by $\hat{\theta}(\theta)$ the solution to (3) with boundary condition $\hat{\theta}(\underline{\theta}) = \theta_1$ and

$$q(\theta) = \begin{cases} q_1(\theta) & \theta \in [\theta_1, \theta_2], \\ q_2(\theta) & \theta \in (\theta_2, \overline{\theta}]. \end{cases}$$

The function $\hat{\theta}(\cdot) : [\underline{\theta}, \overline{\theta}] \to [\underline{\theta}, \overline{\theta}]$ describes the binding incentive constraints in the constrained optimal mechanism. That is, type θ is indifferent between reporting his type truthfully and reporting type $\hat{\theta}(\theta)$, but strictly prefers either to any other report. See Section 4.3 for a derivation and a discussion of the properties of $\hat{\theta}(\cdot)$.

Theorem 1. *For a given $\underline{x} \in (0,1)$, the quantities and inspection probabilities in the optimal mechanism $\mathbb{M}_{\underline{x}}$ are unique. Suppose the parameters are such that the inspection probability $x(\cdot)$ is weakly increasing. Then the quantities and inspection probability follow a three-regime policy: there exist two thresholds[14] θ_1 and θ_2 with $\underline{\theta} < \theta_1 < \theta_2 \leq \overline{\theta}$ such that*

[14] The threshold θ_1 determines the threshold θ_2 uniquely through equation (4).

1. *All types produce the first-best quantity after inspection and truth-telling:* $\forall\theta, q^I(\theta) = q^{FB}(\theta)$

2. - *Low-cost types produce the first-best quantity when not inspected:* $\forall\theta \in [\underline{\theta}, \theta_1], q^N(\theta) = q^{FB}(\theta)$;
   - *Intermediate cost types produce less than the first-best quantity when not inspected:* $\forall\theta \in (\theta_1, \theta_2], q^N(\theta) = q_1(\theta) < q^{FB}(\theta)$;
   - *High-cost types produce less than the first-best quantity when not inspected:* $\forall\theta \in (\theta_2, \overline{\theta}], q^N(\theta)$ *solves*

$$V\left(q^{FB}(\theta)\right) - \theta q^{FB}(\theta) - \kappa = V\left(q^N(\theta)\right) - q^N(\theta)V'\left(q^N(\theta)\right). \tag{5}$$

3. - *Low and intermediate cost types are inspected with the minimal probability:* $\forall\theta \in [\underline{\theta}, \theta_2], x(\theta) = \underline{x}$;
   - *High-cost types are inspected with probability strictly between* $\underline{x}$ *and* 1*:* $\forall\theta \in (\theta_2, \overline{\theta}]$, $x(\theta)$ *is given by the solution to*

$$-x'(\hat{\theta})q^N(\hat{\theta})(\hat{\theta} - \theta) + (1 - x(\hat{\theta}))\frac{\partial q^N}{\partial\hat{\theta}}(\hat{\theta})(\hat{\theta} - \theta) + (1 - x(\hat{\theta}))q^N(\hat{\theta}) = 0, \tag{6}$$

   with the boundary condition $x(\theta_2) = \underline{x}$ *where* $\hat{\theta} = \hat{\theta}(\theta)$.

Proof in Appendix A.2.

Theorem 1 describes the quantities and inspection probabilities in a constrained optimal mechanism for a fixed lower bound $\underline{x} > 0$ on the inspection probability. Together with the transfer scheme described in Proposition 1, Theorem 1 characterizes the optimal constrained mechanism. The Theorem holds for primitives of the model for which the inspection probability is increasing in the optimal mechanism. We provide sufficient conditions for this in Section 4.4. Example 1 satisfies these conditions. We revisit the issue of monotone inspection policy below.

The following example illustrates the optimal constrained mechanism.

Example 1 (continued)**.** *Suppose* $\kappa = \ln(3/2)$ *and* $\underline{x} = 0.1$*. The first-best quantity equals* $q^{FB}(\theta) = 1/\theta$*, and the quantity without inspection given in* (5) *satisfies* $q^N(\theta) = e^{-\kappa}/\theta$*.*

Then the differential equation characterizing the binding incentive constraints, (3) *, for* $x(\theta) > \underline{x}$ *reads*

$$(\hat{\theta}(\theta) - \theta)f(\theta) = \hat{\theta}'(\theta)\left(V'(q^N(\hat{\theta}(\theta))) - \hat{\theta}(\theta)\right)f(\hat{\theta}(\theta)) = \hat{\theta}'(\theta)\hat{\theta}(\theta)\left(e^{\kappa} - 1\right)f(\hat{\theta}(\theta)),$$

with end point condition $\hat{\theta}(\overline{\theta}) = \overline{\theta}$*.*[15] *The unique solution is*

$$\hat{\theta}(\theta) = 2\theta - \frac{\theta^2}{2}.$$

[15] This follows from Lemma 1 and the fact that inspection probabilities are less than 1..

Define $y(\theta) = 1 - x(\theta)$. *By* (6), *we know*

$$-\frac{\mathrm{d}\ln(y(\hat{\theta}))}{\mathrm{d}\hat{\theta}} = \frac{\partial\ \ln\left(\frac{\hat{\theta}-\theta}{\hat{\theta}}\right)}{\partial\ \hat{\theta}}.$$

From the differential equation for $\hat{\theta}(\theta)$, *we know* $\frac{1}{\theta(e^{\kappa}-1)} = \frac{\hat{\theta}'(\theta)}{\hat{\theta}(\theta)-\theta}$. *Therefore,*

$$x(\theta) = 1 - \left(1 - x(\overline{\theta})\right)\frac{2\theta}{(2-\sqrt{4-2\theta})^2}.$$

For [16] $\theta_1 = 1.38$, $\theta_2 = 1.64$. *This yields* $x(\overline{\theta}) = 0.636$. *Consequently, the inspection probability is:*

$$x(\theta) = \begin{cases} 0.1, & \text{for } \theta \in [1, 1.64], \\ 1 - 0.364\frac{2\theta}{(2-\sqrt{4-2\theta})^2}, & \text{for } \theta \in (1.64, 2]. \end{cases}$$

Now we derive $\hat{\theta}(\cdot)$ *and* $q^N(\cdot)$ *for* $\theta \leq \theta_2$. *For ease of computation, we solve for* $h(\cdot) = \hat{\theta}^{-1}(\cdot)$ *instead of* $\hat{\theta}(\cdot)$. *For* $q(\cdot) = q^N(\cdot)$, *the system of differential equations for* $\theta \leq \theta_2$ *is given as*

$$h'(\theta) = \frac{(1/q(\theta) - \theta)h(\theta)}{(\theta - h(\theta))\theta}, \qquad q'(\theta) = -\frac{q(\theta)}{\theta - h(\theta)}.$$

The initial and end point conditions are:

$$h(\theta_2) = 2 - \sqrt{4 - 2\theta_2}, \quad q(\theta_2) = \frac{e^{-\kappa}}{\theta_2}, \quad h(\theta_1) = \underline{\theta}.$$

Solving the above differential equation (numerically) we get $h(\cdot)$ *and* $q(\cdot)$. *Figure 2 shows the quantity without inspection and the inspection policy. Figure 1 shows the function* $\hat{\theta}(\cdot)$. θ_1 *is the lowest type any other type wants to mimic. Higher types want to mimic higher types.*

The first statement of Theorem 1 says that the quantity after inspection and truth-telling is the first-best quantity, $q^I(\theta) = q^{FB}(\theta)$. If the quantity after inspection is not the first-best quantity, i.e., $q^I(\theta) \neq q^{FB}(\theta)$ for some type θ, the Principal can increase her payoff. To see this, note that $q^I(\theta)$ affects incentives only for type θ. Therefore, changing $q^I(\theta)$ to $q^{FB}(\theta)$ and making a compensatory change in $t^I(\theta)$ by $\theta(q^{FB}(\theta) - q^I(\theta))$ leaves the incentive constraints unaffected. Clearly, this change does not violate the (IR-I) constraint either. However, the Principal's payoff increases. This property is intuitive. The rent that has to be paid to the Agent to induce truth-telling is not affected by the quantity after inspection. Therefore, distorting the quantity from its first-best level only reduces the Principal's payoff.

[16] This is the unique (numerically derived) threshold θ_1 such that all conditions in Theorem 1 are satisfied.

Quantities without inspection and the inspection probability are characterized by a three-regime policy. There exist two unique thresholds, θ_1 and θ_2, that divide the type space into three regions: low-cost types $[\underline{\theta}, \theta_1]$; intermediate cost types $(\theta_1, \theta_2]$; and high-cost types $(\theta_2, \overline{\theta}]$.

Low-cost types produce the first-best quantity even when not inspected and are inspected with the minimal probability $\underline{x}$. The property that quantities are not distorted for a positive mass of types follows from two properties. First, incentive constraints only bind upwards. Second, incentive constraints do not bind locally, as argued in Section 4.1. Hence, there must be an interval of types which no other type wants to mimic. Distorting quantities is not optimal, however, when doing so does not decrease information rents. Consequently, there is an interval of low cost types that produce the first-best quantity even when not inspected. Similarly, raising the inspection probability for types no other type wants to mimic above $\underline{x}$ does not reduce information rents, and therefore cannot be optimal.[17]

More precisely, from the reduced problem that $q^N(\theta) \leq q^{FB}(\theta)$ for all types θ. Now, type $\underline{\theta}$ receives the highest information rent. It is not optimal for type $\underline{\theta}$ to mimic any type $\hat{\theta}$ with $(1 - \underline{x})q^{FB}(\hat{\theta})(\hat{\theta} - \underline{\theta}) < \pi(\underline{\theta})$. In particular, type $\underline{\theta}$ does not want to mimic types $\hat{\theta}$ that are close to $\underline{\theta}$. Because the information rent is continuous and decreasing in the type, a similar argument holds for types close enough to $\underline{\theta}$. Consequently, there exists a type θ_1 such that no other type finds it optimal to mimic a type $\hat{\theta} < \theta_1$.

The result regarding the quantities is reminiscent of the familiar "no-distortion-at-the-top" property. However, it is a stronger statement, to the extent that there is a positive mass of types $[\underline{\theta}, \theta_1)$ that produce the first-best quantity when not inspected.

Theorem 1 says that intermediate-cost types are inspected with the minimal inspection probability $\underline{x}$. However, these types produce a quantity less than the first-best quantity when not inspected. For types above the threshold θ_1 there exists a type θ that is indifferent between reporting his true type and mimicking some type in $[\theta_1, \overline{\theta})$.[18] Distorting quantities without inspection downwards and increasing the probability of inspection are substitutes for reducing information rents: both higher inspection probabilities $x(\cdot)$ and lower quantities without inspection, $q^N(\cdot)$, reduce (weakly) the information rents $\pi(\cdot)$. Consequently, the Principal can decrease the information rent she pays by reducing the quantity without inspection or by increasing the inspection probabilities for such types. For intermediate cost types, it is optimal to distort the quantities downwards from first-best instead of increasing the inspection probability. The reason is that the marginal cost of raising the inspection probability, $\kappa > 0$, is larger than the Principal's loss from distorting the quantity downwards from first-best when the quantity distortion is small enough. Hence, types in that range produce less than the first-best quantity.

To see why the quantity without inspection must obey equation (2), note that type

[17] In a tax-audit setting with finitely many income levels, Mookherjee and Png (1989, Lemma 3) find that each report which no other type wants to mimic is inspected with probability 0. However, they do not show that there is a positive mass of types other than the highest type that are inspected with probability 0.

[18] This follows from Lemma 1. We return to this when showing how we deal with the non-locally binding IC constraints in Section 4.3.

θ's payoff when reporting $\theta' \neq \theta$ is $(1-x(\theta'))q^N(\theta')(\theta'-\theta)$. Since θ is indifferent between reporting truthfully and reporting $\hat{\theta}(\theta)$, $\hat{\theta}(\theta)$ must maximize this expression. Equation (6) is the FOC for this maximization problem.[19] We remark that the quantity without inspection in this region depends on the parameters of the model through the binding incentive constraints $\hat{\theta}(\cdot)$.

When the distortion away from first-best in the quantity without inspection becomes large enough, the Principal wants to reduce information rents by inspecting reports with a probability above the lower bound as well. Consequently, for high-cost types, the inspection probability is above the lower bound $\underline{x}$ and quantities without inspection are strictly below the first-best quantity.

To see why quantities without inspection for high-cost types obey equation (5), note that information rents depend on quantities and inspection probability only through $Q^N(\theta) = (1-x(\theta))q^N(\theta)$. In particular, information rents are unaffected if the quantity without inspection is is altered to $\beta q^N(\theta)$ and inspection probabilities are changed to $(1-x(\theta))/\beta$ for a β close to 1. Increasing $q^N(\theta)$ by a factor $\beta \approx 1$ raises the Principal's payoff by

$$V'(q^N(\theta))q^N(\theta) - \theta q^N(\theta).$$

The corresponding change in the inspection probability reduces the Principal's payoff by

$$V(q^{FB}(\theta)) - \theta q^{FB}(\theta) - \kappa - \left(V(q^N(\theta)) - \theta q^N(\theta)\right).$$

In the constrained optimal mechanism, such a change cannot increase the Principal's payoff. Such a change is feasible whenever $\underline{x} < x(\theta) < 1$. Consequently, when the inspection probability is interior in the constrained optimal mechanism, both effects must offset each other: the quantity without inspection must satisfy (5).[20] The quantity without inspection $q^N(\theta)$ given by equation (5) is strictly less than the type's first-best quantity, $q^{FB}(\theta)$, for any $\kappa > 0$.[21]

[19] The assumption that there is no interval (θ', θ'') such that $q^{FB}(\theta) = 1/(c_1\theta - c_2)$ for $\theta \in (\theta', \theta'')$ for positive constants $c_1 > 0, c_2 \in [\underline{\theta}, \overline{\theta}]$ ensures that $\theta q^{FB}(\theta)$ is not constant on any interval if positive length. As a consequence, $q^{FB}(\cdot)$ is not a solution to (2).

[20] Equation (5) implies that the quantity without inspection does not depend on the distribution of types when $x(\theta) > \underline{x}$. The set of types for which $x(\theta) > \underline{x}$ depends on the distribution of types and the other parameters of the model. The expected quantity without inspection, $(1-x(\theta))q^N(\theta)$, depends on the distribution of types. The reason for this is as follows. Information rents depend only on $(1-x(\cdot))q^N(\cdot)$. Hence, one can maximize the Principal's payoff locally by changing $x(\theta)$ and $q^N(\theta)$, leaving $(1-x(\theta))q^N(\theta)$ constant. As this change affects the objective for a single type θ only, the density at this type $f(\theta)$ does not affect the trade-off between a lower quantity and higher inspection probability for θ. Theorem 1 also implies that a change in the distribution of types that affects the inspection probability in the constrained optimal mechanism does not change the quantity without inspection for types that are inspected with probability $x(\theta) > \underline{x}$ before and after the change.

[21] There are parameters such that $q^{MR}(\theta) > q^N(\theta)$, that is, for some types the quantity without inspection is distorted more than in the setting where inspection is not possible. For example, let $V(q) = \ln(q)$ so that $q^N(\theta) = e^{-\kappa}/\theta$ for $x(\theta) > \underline{x}$. Assuming $[\underline{\theta}, \overline{\theta}] = [1, 2]$ and a uniform distribution over types, the optimal quantity is $q^{MR}(\theta) = \frac{1}{\theta(1+\ln(\theta))}$. If the cost of inspection, κ, is large enough, $e^{\kappa} > 1 + \ln(\theta)$ so that $q^{MR}(\theta) > q^N(\theta)$.

The qualitative features of the quantity allocation of Theorem 1 differ from the optimal quantities in related papers. First, Baron and Besanko's (1984) seminal "separation" result fails. Baron and Besanko (1984) find that quantity distortions are independent of the inspection cost. Moreover, the optimal quantities equal the optimal quantities in a setting where inspection is not feasible. This separation does not obtain in our model. In Palonen and Pekkarinen (2022), the set of types for which quantities are distorted downward from first-best depends on the inspection cost. However, the size of the distortion is independent of the inspection cost. In our model, both the size of the distortion and the set of types for which quantities are distorted depend on the inspection cost. Second, quantities in Baron and Besanko (1984) are inefficiently low for all but a single type — even for types that are inspected with probability 1. Theorem 1 shows that this is false in our model. Quantities are not distorted for a non-trivial interval of types.[22] For this interval of types, inspection occurs with the minimal probability $\underline{x}$. Quantity distortions for all types are thus not optimal.

Finally, Theorem 1 pins down the inspection probabilities for high-cost types. High-cost types are inspected with probability strictly between the lower bound $\underline{x}$ and 1.

The intuition why no type is inspected with probability 1 is the following. Suppose some types $\Theta_1 \subset (\underline{\theta}, \overline{\theta})$ are inspected with probability 1 but some higher type θ' is inspected with probability less than 1. No type wants to report a type that is inspected with probability 1 as doing so would yield a payoff of 0. Instead, types θ below θ' could report θ' instead, securing a positive payoff in the event that the report is not inspected. Hence, one could decrease the probability with which types in Θ_1 are inspected slightly and have them produce the first-best quantity when not inspected. As long as the decrease in the inspection probability is small enough, no type wants to mimic a type in Θ_1. This change in the mechanism saves inspection costs while leaving information rents unchanged. Clearly, this cannot occur in the optimal constrained mechanism. It remains to rule out that there is no threshold above which all types are inspected with probability 1. Suppose there is a cut-off $\theta^\dagger$ such that all types above $\theta^\dagger$ are inspected with probability 1. Note that inspecting all types above $\theta^\dagger$ cannot be optimal if it was not for the effect on the information rents for type θ less than $\theta^\dagger$. The Principal can thus reduce the inspection probability for types above $\theta^\dagger$ slightly and increase her payoff from these types. This increases information rents for types below $\theta^\dagger$. The effect on information rents, however, is of second-order importance: as long as the reduction in inspection probability is small enough, the Principal benefits.[23] Consequently, no type is inspected with probability 1 in the optimal constrained mechanism.

Mookherjee and Png (1989) show a similar result: no type is inspected with probability 1. In contrast, the optimal inspection policy in Baron and Besanko (1984) has a

[22] In Palonen and Pekkarinen (2022), there is a non-empty interval of types for which quantities are first-best even when not inspected. However, the reasons in their and our setting differ. Their result obtains because both the Principal's and the Agent's preferences are linear in quantities. Their result does not obtain if the Principal's preferences over quantities are strictly concave. In our setting, quantities for an interval of types are not distorted for a different reason: local incentive constraints do not bind.

[23] See the proof of Claim A.2.7 for the precise argument.

"bang-bang"-property: types below a threshold are not inspected while all types above this threshold are inspected with probability 1. Similarly, in Palonen and Pekkarinen (2022), a report is either not inspected, or the inspection probability for reports that are inspected with positive probability is constant. Whether this probability equals 1 depends on parameters.

Knowing the optimal quantities, $q^N(\cdot)$, and the binding incentive constraints, $\hat{\theta}(\cdot)$, we can recover the inspection probability, $x(\cdot)$, in a constrained optimal mechanism. Note that type θ's payoff when reporting $\theta' \neq \theta$ is $(1 - x(\theta'))q^N(\theta')(\theta' - \theta)$. Since θ is indifferent between reporting truthfully and reporting $\hat{\theta}(\theta)$, $\hat{\theta}(\theta)$ must maximize this expression. Equation (6) is the FOC for this maximization problem.[24]

Theorem 1 says that the quantities and inspection probabilities in the constrained optimal mechanism are unique. Transfers, however, are not uniquely pinned down. Proposition 2 delineates the set of transfers that feature in any constrained optimal mechanism.

Proposition 2. *Let $(x(\cdot), q^I(\cdot), q^N(\cdot))$ be as in Theorem 1. Then the mechanism is constrained optimal if and only if*

- *for $\theta \leq \theta_1$, $t^N(\theta) \in [\theta q^N(\theta), \frac{\pi(\theta)}{1-\underline{x}} + \underline{\theta} q^{FB}(\theta)]$ and $\underline{x} t^I(\theta) + (1 - \underline{x}) t^N(\theta) = \pi(\theta) + \theta q^{FB}(\theta)$;*
- *for $\theta > \theta_1$, $t^N(\theta) = \theta q^N(\theta)$ and $t^I(\theta)$ is given by (1).*

Moreover, $t^I(\theta) > \theta q^I(\theta)$, i.e., (IR-I) is slack after inspection and truthful reporting, for all types θ.
Proof in Appendix A.3.

Proposition 2 states that for intermediate and high-cost types, the transfer without inspection is given by $t^N(\theta) = \theta q^N(\theta)$, i.e., (IR-N) binds. To see why this holds, suppose to the contrary that some type θ gets paid more than the cost of production, $t^N(\theta) > \theta q^N(\theta)$. Then the Principal can raise the quantity without inspection and simultaneously increase the transfer after inspection $t^I(\theta)$ to offset the increase in production cost. Since the transfer without inspection does not change, the incentives for all other types $\theta' \neq \theta$ are preserved as well. Moreover, (IR-N) remains slack. The Principal benefits from the increase in the quantity as long as the original quantity is strictly below first-best. Hence, only for types that produce the first-best quantity when not inspected can (IR-N) be slack. By Theorem 1, these are the low-cost types $\theta \leq \theta_1$.

Proposition 2 further states that, for low-cost types, the transfer without inspection can be strictly above the production cost. Recall that, for all types θ, reporting a type θ' below $\underline{\theta}$ yields a payoff strictly below the payoff from reporting truthfully, when $t^N(\theta') = \theta' q^N(\theta')$. This remains true if such types θ' get a slightly higher transfer. Thus, there exists a maximal transfer without inspection, $\overline{t^N}(\theta)$, such that for all lower

[24] Our methods allow us to derive the binding incentive constraints $\hat{\theta}(\cdot)$ without knowledge of the optimal inspection policy $x(\cdot)$. This allows us to back out the inspection policy using $\hat{\theta}(\cdot)$ and $q^N(\cdot)$. See Section 4.3 and in particular Lemma 2.

transfers without inspection, incentives for all types remain unchanged. The transfer with inspection, $t^I(\theta)$, is changed to keep the expected transfers from truth-telling equal to the expected transfers when (IR-N) binds and (1) holds. If a type $\theta' \leq \theta_1$ gets a transfer $t^N(\theta') > \overline{t^N}(\theta')$, some type θ strictly prefers reporting θ' to reporting truthfully.

The maximal transfer without inspection $\overline{t^N}(\theta)$ is decreasing in θ. This holds because higher types produce a lower quantity when not inspected so that the production cost when reporting a higher type is decreasing. Consequently, the maximal transfer must be decreasing to preserve incentives.

Proposition 2 also says that all types get a bonus payment when inspected and found to have reported truthfully: (IR-I) is slack for all types and truthful reports. Bonus payments are thus a feature of all constrained optimal mechanisms.

A comment on the hypothesis of increasing inspection probability in Theorem 1 is in order. The hypothesis is stronger than needed. It is sufficient that the inspection probability is strictly greater than the minimal probability, i.e., $x(\theta) > \underline{x}$, for types above the threshold θ_2. Even if that is not the case, we can characterize the optimal mechanism: the regimes for quantity and inspection probability between θ_1 and θ_2, and between θ_2 and $\overline{\theta}$ alternate. More precisely, there is a sequence of thresholds $(\theta_i)_{i=1}^n$ such that the quantity without inspection is given as the solution to (2) and $x(\theta) = \underline{x}$ for $\theta \in (\theta_{2k-1}, \theta_{2k})$, whereas the quantity without inspection is given by equation (5) for $\theta \in [\theta_{2k}, \theta_{2k+1}]$.

Theorem 1 is mute on the existence of a region of types that are inspected with probability above the minimal inspection probability, $x(\theta) > \underline{x}$. In fact, there are parameters such that no type is inspected with probability greater than $\underline{x}$. This happens when the cost of inspection, κ, is large. More precisely, there exists a $\underline{\kappa} > 0$ such that for all $\kappa \geq \underline{\kappa}$, all types θ are inspected with the minimal probability $\underline{x}$ in any constrained optimal mechanism. To see this, note that the solution to equation (5) converges to 0 as $\kappa \to \infty$. Since $V'(q) \to_{q \to 0} \infty$, it cannot happen that a positive mass of types is inspected with probability strictly above $\underline{x}$ when κ is large enough. Consequently, all types are inspected with probability $\underline{x}$ and quantities are independent of the inspection cost for all κ above a threshold. These quantities do not coincide with the optimal quantities in a setting where inspection is infeasible: the optimal constrained mechanism does *not* converge to the optimal mechanism when inspection is infeasible even as the inspection cost κ grows large![25]

If the inspection cost κ is small enough, the threshold θ_2 is interior. Intuitively, when the inspection cost is small enough, it is optimal to inspect some reports with probability strictly above the minimal probability $\underline{x}$. To see this, assume toward a contradiction that $\theta_2 = \overline{\theta}$ for all κ small enough. Note that the value of the reduced problem converges to the first-best value as κ goes to 0. If all types are inspected with the minimal probability $\underline{x}$, they produce a quantity when not inspected $q^N(\theta) \geq q_2(\theta)$ where $q_2(\theta)$ solves equation (5). As the inspection cost vanishes, $q_2(\theta)$ approaches the first-best quantity. The Principal's payoff, in the limit, when inspecting all types with

[25] We remark that this is not due to the lower bound on the inspection probability; see Lemma 4 and Theorem 2.

the minimal probability is then

$$\int \left\{ V(q^{FB}(\theta)) - \theta q^{FB}(\theta) - \sup_{\hat{\theta}} (1 - \underline{x}) q^{FB}(\hat{\theta})(\hat{\theta} - \theta) \right\} \mathrm{d}F(\theta).$$

This is strictly less than the first-best payoff minus the cost of inspecting all types with probability 1 for small enough κ, a contradiction. Hence, the threshold θ_2 is interior when the inspection cost κ is small enough.

The existence of a region in which types are inspected with probability $\underline{x}$ but produce strictly less than their first-best quantity is guaranteed by Theorem 1, for all parameters. This implies that the constrained optimal contract does not implement the first-best quantity with probability 1.

4.3. Global incentive compatibility

In this subsection, we deal with the main technical challenge we face when deriving the constrained optimal mechanism: global incentive compatibility. We argue in Section 4.1 that incentive constraints do not bind locally, but only globally. One major advantage of locally binding incentive compatibility constraints is not the fact that the constraints bind locally but that we know *which* constraints bind. Nonetheless, we manage to characterize which incentive constraints bind.

Define the correspondence[26]

$$\hat{\theta}(\theta) = \arg\max_{\hat{\theta}} \ (1 - x(\hat{\theta})) q^N(\hat{\theta})(\hat{\theta} - \theta).$$

The correspondence gives the binding incentive constraints for type θ; that is, for any $\hat{\theta} \in \hat{\theta}(\theta)$, the Agent is indifferent between reporting his true type θ and type $\hat{\theta}$ because, by our choice of $t^I(\theta)$ through (1),

$$x(\theta)(-q^{FB}(\theta)\theta + t^I(\theta)) = (1 - x(\hat{\theta})) q^N(\hat{\theta})(\hat{\theta} - \theta).$$

The correspondence $\hat{\theta}(\cdot)$ depends on the entire functions $q^N(\cdot)$ and $x(\cdot)$. Moreover, it is a priori not clear that the correspondence is well-behaved in an optimal mechanism. In particular, $\hat{\theta}(\cdot)$ may be empty- or multi-valued. Nonetheless, we can show the following Lemma.

Lemma 1. *The solution to the constrained problem is such that*

1. $(1 - x(\cdot))q^N(\cdot)$ *is a differentiable function that is strictly decreasing when positive;*

[26] When local incentive constraints bind, $\hat{\theta}(\theta) = \{\theta\}$. In particular, when local incentive constraints are sufficient and necessary for global incentive compatibility, $\hat{\theta}(\cdot)$ does not depend on the quantity. When the allocation is monotone in the type, transfers exist such that the incentive constraints hold, even without inspection. Consequently, standard optimal control techniques can be applied. This is not possible in our case: not only are the incentive constraints not locally binding, but the set of binding constraints depends on the entire functions $x(\cdot)$ and $q^N(\cdot)$.

2. $\hat{\theta}(\cdot)$ is single-valued and increasing.

Proof in Appendix A.4

The intuition behind the continuity and monotonicity of $(1 - x(\cdot))q^N(\cdot) \equiv Q^N(\cdot)$ is the following. Suppose that $Q^N(\cdot)$ is weakly increasing on the interval $(\theta', \theta^\dagger)$. Then $Q^N(\theta^\dagger)(\theta^\dagger - \theta) > Q^N(\hat{\theta})(\hat{\theta} - \theta)$ for all $\hat{\theta} \in (\theta', \theta^\dagger)$ and for every type $\theta \leq \theta'$. Consequently, no type wants to mimic any such $\hat{\theta}$ and the Principal can receive a higher payoff by increasing $q^N(\cdot)$ on the interval $(\theta', \theta^\dagger)$.

Continuity of $Q^N(\cdot)$ holds for a similar reason. Suppose $Q^N(\cdot)$ has a downward jump at θ'. Since $Q^N(\cdot)$ is decreasing, no type wants to mimic any $\hat{\theta} \in (\theta', \theta' + \delta)$ for some small but positive δ. Consequently, the Principal can increase her payoff by increasing $q^N(\cdot)$ on $(\theta', \theta' + \delta)$.

Continuity of $Q^N(\theta)$ implies that $\hat{\theta}(\cdot)$ is not empty–valued. The other statements in the Lemma are tedious to prove. See the proof for details.

Recall that, in a model where inspection is not possible, monotonicity of $q^N(\cdot)$ is necessary and sufficient for the existence of a transfer scheme $t^N(\cdot)$ that implements $q^N(\cdot)$ in an incentive compatible manner. However, when inspection is feasible, monotonicity of $q^N(\cdot)$ or $Q^N(\cdot)$ is not required for incentive compatibility; any function $q^N(\cdot)$ can be implemented in an incentive compatible mechanism, however weirdly behaved. The monotonicity result in Lemma 1 is due not to the *feasibility*, but to the *optimality* of implementing an allocation $Q^N(\cdot)$. Allocations for which $Q^N(\cdot)$ is not strictly decreasing are not optimal in the constrained problem because they entail types no other type wants to mimic, and consequently, for which quantity distortions or inspection with probability above the lower bound cannot be optimal.

Lemma 1 shows that the set of binding incentive constraints is well-behaved. Unfortunately, this is not enough to solve for the constrained optimal mechanism: we also need to know which constraints bind. Yet, we are able to characterize the set of binding incentive constraints.

Lemma 2. *In the constrained optimal mechanism, for all $\theta \in \left(\hat{\theta}(\underline{\theta}), \overline{\theta}\right)$ such that $\pi(\theta) > 0$, the function $\hat{\theta}(\cdot)$ is strictly increasing, differentiable and obeys the differential equation (3), i.e.,*

$$(\hat{\theta}(\theta) - \theta) f(\theta) = \hat{\theta}'(\theta) \left(V'(q^N(\hat{\theta}(\theta))) - \hat{\theta}(\theta) \right) f(\hat{\theta}(\theta))$$

with an appropriate boundary condition.
Proof in Appendix A.5.

The intuition behind this result is closely related to results in mechanism design without inspection. Recall that, under regularity conditions, the optimal quantity in a setting without inspection solves

$$V'(q^{MR}(\theta)) = \theta + \frac{F(\theta)}{f(\theta)}.$$

Multiplying both sides by $q^{MR}(\theta)$, rearranging and integrating over $[\underline{\theta}, \overline{\theta}]$ yields

$$\int_{[\underline{\theta},\overline{\theta}]} q^{MR}(\theta)[V'(q^{MR}(\theta)) - \theta]\,\mathrm{d}F(\theta) = \int_{[\underline{\theta},\overline{\theta}]} q^{MR}(\theta)\frac{F(\theta)}{f(\theta)}\,\mathrm{d}F(\theta).$$

The left-hand side is a weighted measure of the distortion of the second-best quantity $q^{MR}(\theta)$ away from the first-best quantity. The information rent π^{MR} in the problem without inspection solves $\dot{\pi}^{MR}(\theta) = -q^{MR}(\theta)$ and hence

$$\int_{[\underline{\theta},\overline{\theta}]} q^{MR}(\theta)\frac{F(\theta)}{f(\theta)}\,\mathrm{d}F(\theta) = \int_{[\underline{\theta},\overline{\theta}]} \pi^{MR}(\theta)\,\mathrm{d}F(\theta).$$

The equation above states that the distortion needs to equal the information rent.

Similarly, we can show that, in our setting with inspection, for each interval $I \subset [\underline{\theta}, \overline{\theta}]$

$$\int_{\{\theta\in I\}} (1 - x(\theta))q^{N}(\theta)[V'(q^{N}(\theta)) - \theta]\,\mathrm{d}F(\theta) = \int_{\{\theta' : \hat{\theta}(\theta')\in I\}} \pi(\theta')\,\mathrm{d}F(\theta').$$

The difference to the case without inspection is that the integration of the information rent is over the set $\hat{\theta}(\theta) \in I$ to account for the fact that local incentive constraints do not bind. The differential equation for $\hat{\theta}(\cdot)$ in Lemma 2 follows from the equation in the last display.[27]

Lemmas 1 and 2 have another implication. For every type θ there is exactly one type $\hat{\theta} > \theta$ such that type θ is indifferent between reporting his true type and type $\hat{\theta}$. Conversely, for every type $\hat{\theta}$ above the threshold θ_1 there exists exactly one type θ that is indifferent between reporting his true type and the type $\hat{\theta}$.

4.4. Monotonicity of inspection

So far, we have not shown that the probability of inspection is monotone in the Agent's reported type. In fact, this need not be the case; Example 2 shows that the inspection probability can be strictly decreasing in the reported cost for some types. However, we provide sufficient conditions on the primitives of the model that guarantee that the inspection probability is increasing in the Agent's type. Proposition 3 states the sufficient conditions.

Proposition 3. *Let $q(\cdot)$ be given as in equation (5). The probability of inspection in the constrained optimal mechanism is weakly increasing if one of the following three conditions holds:*

1. *For all $\theta \in [\underline{\theta}, \overline{\theta}]$, $\theta \mapsto \theta q(\theta)$ is a weakly increasing function.*

[27] The probability of inspection factors out from both sides of the integral equation. Hence, $x(\cdot)$ does not show up in the differential equation (3). The binding incentive constraints $\hat{\theta}(\cdot)$ depend on the inspection policy $x(\cdot)$ through the boundary condition for the differential equation.

2. *The third derivative of the Principal's valuation for quantity is negative,* $V''' \leq 0$, *and the following inequality is true:*

$$\frac{1}{\overline{\theta} - \underline{\theta}} \geq -\frac{q'(\underline{\theta})}{q(\underline{\theta})} = -\frac{q^{FB}(\underline{\theta})}{q^2(\underline{\theta})V''(q(\underline{\theta}))}.$$

3. *The Principal's preferences over quantities are of the CRRA type,* $V(q) = q^{1-\alpha}/(1-\alpha)$, *for* $\alpha \geq 1$.

Proof in Appendix A.10.

The conditions in Proposition 3 are straightforward to verify. The idea behind the proof of Proposition 3 is to solve for the slope of the inspection probability using (6). One then provides a lower bound on the slope of the inspection probability. The conditions in Proposition 3 guarantee that the lower bound is positive.

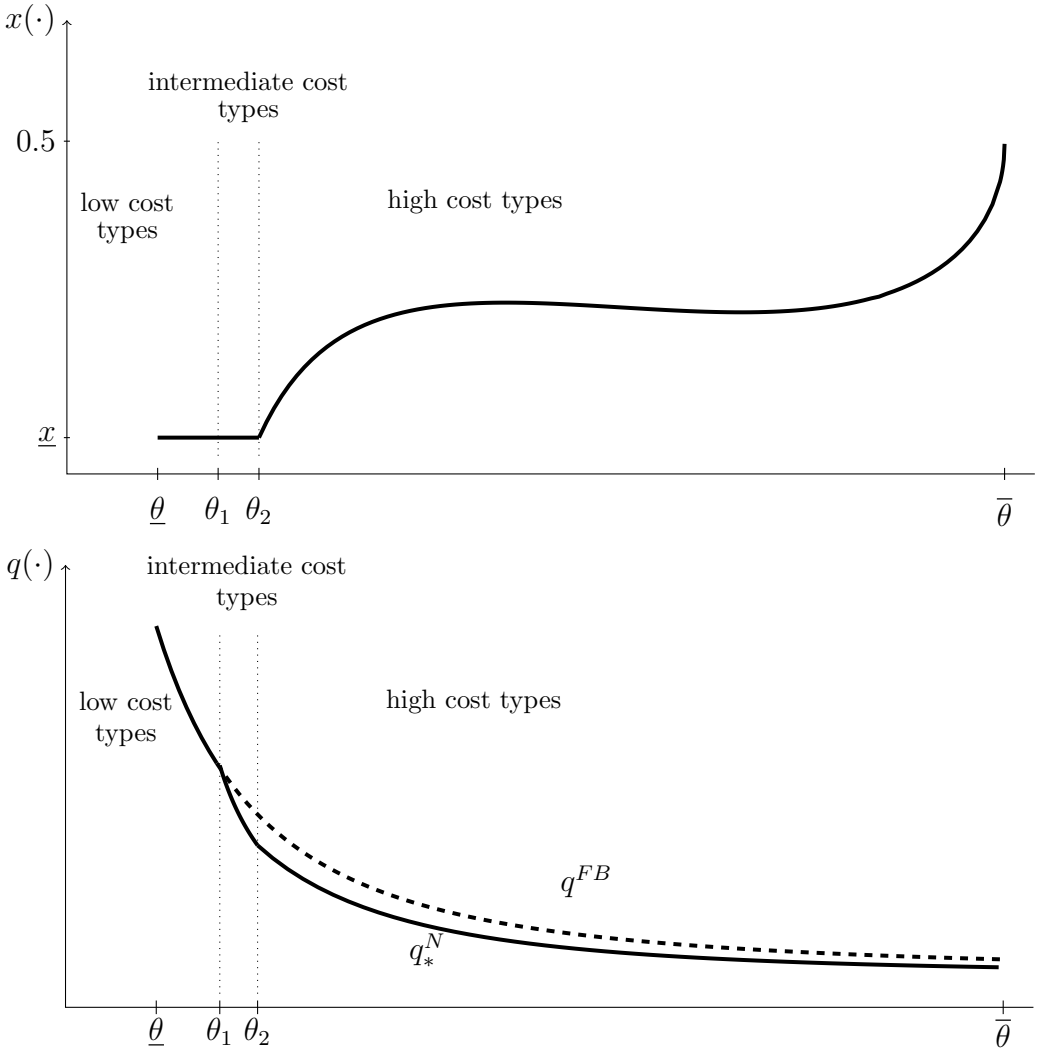
The lower bounds used in deriving the conditions for Proposition 3 do not rely on solving for $\hat{\theta}(\cdot)$ explicitly, but hold when $\theta \leq \hat{\theta}(\theta) \leq \overline{\theta}$ for all types θ. This is the reason why the conditions in Proposition 3 do not depend on the distribution of types f.

Proposition 3 provides sufficient conditions for the inspection probability to be increasing in the Agent's reported type. Example 2 exhibits parameter values such that the inspection probability is strictly decreasing for some type.

In Palonen and Pekkarinen (2022), the inspection probability is not monotone in the reported types; however, inspection probabilities are monotone for all types that are allocated a positive quantity.[28] Inspection probabilities are monotone in Border and Sobel (1987) and Chander and Wilde (1998), as well as in Baron and Besanko (1984) for all parameters such that each type of the Agent receives a payoff strictly above the outside option. Monotonicity fails in Mookherjee and Png (1989).

Example 2. *This example shows that the optimal probability of inspection is not increasing in the reported cost. Assume* $V(q) = 2\sqrt{q}$, $\kappa = 0.0495$, $[\underline{\theta}, \overline{\theta}] = [1.4, 7]$, $\underline{x} = 0.01$, *and* $F(\cdot)$ *is the uniform distribution. The first-best quantity is* $q^{FB}(\theta) = \theta^{-2}$. *Because* $\overline{\theta} < 1/\kappa$, *the solution to equation (5) is given by* $q(\theta) = (1/\theta - \kappa)^2$. *We numerically solve (6) to obtain the inspection probability in the optimal mechanism. The first panel in Figure 3 plots the probability of inspection in the optimal mechanism. The inspection probability is constant and equal to the minimal inspection probability for low and intermediate cost types. For high cost types, the probability of inspection is strictly above* $\underline{x}$. *However, the inspection probability is strictly decreasing on an interval of high cost types. The second panel in Figure 3 plots the quantities in the optimal mechanism for this example.*

[28] In Palonen and Pekkarinen (2022), there exists an "exclusion region", i.e., a set of types that do not get allocated the good and are inspected with probability 0.



This figure shows the inspection probability and quantity allocation in the constrained optimal mechanism for Example 2. The first panel shows the inspection probability. Types are depicted on the horizontal axis. The vertical axis shows the inspection probability $x(\cdot)$. The inspection probability equals the minimal probability of inspection, $\underline{x}$, for low and intermediate cost types $\theta \in [\underline{\theta}, \theta_2]$. The inspection probability increases strictly at the cut-off θ_2, but decreases strictly for higher types. In particular, the probability of inspection is not monotonically increasing in the type.
The second panel shows the quantity allocation in the constrained optimal mechanism for Example 2. Types are depicted on the horizontal axis. The vertical axis shows quantities. The dashed line is the first-best quantity. The solid line shows the quantity without inspection in the optimal mechanism. Low cost types between $\underline{\theta}$ and θ_1 produce the first-best quantity when not inspected. Intermediate and high cost types produce strictly less than the first-best quantity when not inspected.

Figure 3: Quantity and inspection probability in the constrained optimal mechanism for Example 2.

5. Unconstrained Problem

In this section, we study the unconstrained Principal's problem. That is, we drop the constraint that each report has to be inspected with probability at least $\underline{x}$. Instead, the Principal can choose to not inspect some reports. We show that the unconstrained problem has a value, but an optimal mechanism does not exist. We exhibit the limits of ε-optimal mechanisms and show how they relate to the limits (as $\underline{x} \to 0$) of the constrained optimal mechanism.

Let W_{FB} be the first-best payoff, that is, the maximal payoff to the Principal when she knows the Agent's type. The first-best payoff equals

$$W_{FB} = \int_{\underline{\theta}}^{\overline{\theta}} V(q^{FB}(\theta)) - \theta q^{FB}(\theta) \, \mathrm{d}F(\theta).$$

Lemma 3. *1. The unconstrained Principal's problem $\mathcal{P}_0$ has a value W_0 strictly less than the first-best payoff, $W_0 < W_{FB}$. The Principal's problem $\mathcal{P}_0$ does not admit a solution.*

2. Take a sequence of positive numbers $\{\underline{x}_n\}$ that converge to 0. Then the values of the constrained problems converge to the value of the unconstrained problem, i.e.,

$$\lim_{n \to \infty} W_{\underline{x}_n} = W_0.$$

Proof in Appendix A.7

Lemma 3 has two parts. First, the Principal cannot obtain the first-best value. This holds because the Agent can reject the mechanism ex-post. To induce the Agent to not reject the mechanism ex-post, he must receive a non-negative payoff when being inspected and when not being inspected. As a consequence, the total information rents the Agent receive is either bounded away from 0 or all reports are inspected with probability close to 1. In both cases, the cost to the Principal is non-negligible. It is therefore not possible to approximate the first-best allocation at first-best cost.

Lemma 3 also states that there is no optimal mechanism. Intuitively, the Principal can increase her payoff by reducing the probability of inspection $x(\theta)$ for some types θ, thus saving the inspection costs, and simultaneously increasing the transfer after inspection $t^I(\theta)$ while keeping the expectation $x(\theta)t^I(\theta)$ constant so that the incentive constraints continue to hold. Recall from Section 4.1 that it is optimal to provide all incentives for a type that is inspected with positive probability through transfers after inspection and truth-telling, and that this property does not depend on the magnitude of the inspection probability. Hence, inspecting a type θ with positive probability $x(\theta) > 0$ instead of probability 0 can lead to a discontinuous downward jump in the information rents that need to be paid to other types. The size of this discontinuous jump does not depend on the size of the inspection probability $x(\theta)$. The Principal would like to inspect some types with a probability equal to the smallest positive number—which, of course, is an absurdity.

The non-existence of optimal mechanisms in similar models is well known; see, for example, Becker (1968), Stigler (1970), and Mirrlees (1999). In these models, the Principal can punish the Agent for lying or not exerting effort, and there is no limit on punishments. The Principal wants to inspect the Agent with vanishing probability and punish the Agent with increasing harshness when found to have lied or not exerted effort.[29] However, the reason here is different. The Principal cannot punish the Agent harshly for misreporting his type: he is protected by the (IR-I) and (IR-N) constraints. Instead, the Principal can reward the Agent with bonus payments after truthful inspection. Since there is no upper bound on transfers, the Principal wants to increase the transfer and decrease the probability of inspection for some types, holding expected transfers $x(\theta)t^I(\theta)$ constant. This reduces the inspection cost $x(\theta)\kappa$.

The last part of Lemma 3 states that the value of the constrained problem approaches the value of the unconstrained problem as the minimal inspection probability $\underline{x}$ vanishes.

We next state an observation about the minimal inspection probability $\underline{x}$ in the constrained problem. The minimal inspection probability affects the optimal constrained mechanism only through the inspection probability. In particular, $\underline{x}$ does not affect quantities in an optimal constrained mechanism.

Lemma 4. *The quantities in the constrained optimal mechanism do not depend on the minimal inspection probability. Formally, let $\underline{x}, \underline{x}' \in (0,1)$. Then there is a solution $\mathbb{M}_{\underline{x}} = \left(x_{\underline{x}}(\cdot), q^I_{\underline{x}}(\cdot), t^I_{\underline{x}}(\cdot), q^N_{\underline{x}}(\cdot), t^N_{\underline{x}}(\cdot)\right)$ to $\mathcal{P}_{\underline{x}}$ and a solution to $\mathcal{P}_{\underline{x}'}$, $\mathbb{M}_{\underline{x}'} = \left(x_{\underline{x}'}(\cdot), q^I_{\underline{x}'}(\cdot), t^I_{\underline{x}'}(\cdot), q^N_{\underline{x}'}(\cdot), t^N_{\underline{x}'}(\cdot)\right)$, such that*

$$\left(q^I_{\underline{x}'}(\cdot), q^N_{\underline{x}'}(\cdot)\right) = \left(q^I_{\underline{x}}(\cdot), q^N_{\underline{x}}(\cdot)\right).$$

Moreover, the inspection probabilities $x_{\underline{x}}$ and $x_{\underline{x}'}$ are related by

$$\frac{1 - x_{\underline{x}}(\theta)}{1 - \underline{x}} = \frac{1 - x_{\underline{x}'}(\theta)}{1 - \underline{x}'}.$$

Proof in Appendix A.8.

Lemma 4 states that the minimal inspection probability does not affect the quantities in an optimal mechanism. This holds for two reasons. First, the optimal quantity after inspection is independent of the inspection probability and the quantity without inspection. Second, scaling the probability of not inspecting $1 - x(\cdot)$ by the same factor for every type does not change the incentives: the optimal misreport of a type θ, $\hat{\theta}(\theta)$, is the same under the inspection policies $1 - x(\cdot)$ and $\beta(1 - x(\cdot))$. Thus, scaling the probability of not inspecting by the same factor for each type affects information rents in the same way as it affects the trade-off between higher costs of inspection and lower quantities without inspection compared to inspection,

$$\int (1 - x(\theta))(V(q^N(\theta) - \theta q^N(\theta) - (V(q^{FB}(\theta)) - \theta q^{FB}(\theta) - \kappa))\, \mathrm{d}F(\theta).$$

[29] Unlike in our model, the Principal achieves the first-best payoff in these papers.

As a consequence, a change in the constraint $1-x(\cdot) \geq 1-\underline{x}$ scales the optimal inspection policy, $1-x(\cdot)$, and leaves quantities without inspection unchanged.[30]

Quantities and the binding incentive constraints in the constrained optimal mechanism are not affected by the minimal inspection probability. In particular, the thresholds θ_1 and θ_2 do not depend on the lower bound $\underline{x}$. With the exception of the transfer after inspection, $t^I(\cdot)$, all limits exist as the lower bound vanishes. Importantly, for types θ that are inspected with the minimal probability, $x_{\underline{x}}(\theta) = \underline{x}$ ($\iff$ $x_{\underline{x}'}(\theta) = \underline{x}'$), the probability of inspection converges to 0 as the lower bound vanishes. The expected payoff of type θ converges to

$$\pi_0(\theta) = \sup_{\hat{\theta}}(1 - x_0(\hat{\theta}))q^N(\hat{\theta})(\hat{\theta} - \theta) = (1 - x_0(\hat{\theta}(\theta)))q^N(\hat{\theta}(\theta))(\hat{\theta}(\theta) - \theta)$$

where $x_0(\theta)$ denotes the limit of the inspection probability. The last equality holds because changing the lower bound on the inspection probability does not affect the binding incentive constraints. The maximal transfer without inspection, $\overline{t^N}(\theta)$, from Proposition 2, converges to $\pi_0(\underline{\theta}) + \underline{\theta}q^{FB}(\theta)$ for types $\theta \leq \theta_1$. For low and intermediate cost types $\theta \leq \theta_2$, the transfer after inspection, $t^I(\theta)$, grows unboundedly as the minimal inspection probability vanishes, irrespective of how we select transfers. This is closely related to the reason for the non-existence of an optimal mechanism without a lower bound on the inspection probability. However, the expected transfer after inspection, $x(\theta)t^I(\theta)$, has a well-defined limit for every type θ.[31]

The main result of this section, Theorem 2, relates the quantity allocation and inspection policy of the constrained optimal mechanism to the limits (as ε vanishes) of the quantities and inspection probability of all ε-optimal mechanisms. Denote by $\left(q^I_*(\cdot), q^N_*(\cdot)\right)$ the quantities with and without inspection given in Theorem 1. Recall that, by Lemma 4, these quantities do not depend on the lower bound on the inspection probabilities $\underline{x}$.

Theorem 2. *1. There exists an inspection policy $x_*(\cdot)$ with the following property: for every $\varepsilon > 0$, one can find transfers $(t^I_\varepsilon(\cdot), t^N_\varepsilon(\cdot))$ and a number $g_\varepsilon > 0$ such that*

$$\mathbb{M}_\varepsilon = \left(x_\varepsilon(\cdot) = 1 - g_\varepsilon(1 - x_*(\cdot)), q^I_*(\cdot), t^I_\varepsilon(\cdot), q^N_*(\cdot), t^N_\varepsilon(\cdot)\right)$$

is an ε-optimal mechanism. Moreover, $\lim_{\varepsilon\to 0} g_\varepsilon = 1$.

2. Let $\mathbb{M}_n$ be a sequence of $1/n$-optimal mechanisms for $\mathcal{P}_0$, with quantities $(q^I_n(\theta), q^N_n(\theta))$ and inspection probability $x_n(\theta)$ that converge point-wise almost everywhere. Then

$$\lim_{n\to\infty}(x_n(\theta), q^I_n(\theta), q^N_n(\theta)) = (x_*(\theta), q^I_*(\theta), q^N_*(\theta)).$$

Proof in Appendix A.9.

[30] Observe that the reduced problem is affine in the entire function $1 - x(\cdot)$. Note, however, that the problem is *not* affine in the probability of inspecting a single type, $x(\theta)$.

[31] To be precise, the following is true: for a given $\underline{x}$, $x(\theta)t^I(\theta) \in [b_{\underline{x}}(\theta), B_{\underline{x}}(\theta)]$ for all constrained optimal mechanisms for types $\theta \leq \theta_1$. The limits of $b_{\underline{x}}(\theta)$ and $B_{\underline{x}}(\theta)$ as $\underline{x}$ vanishes are finite.

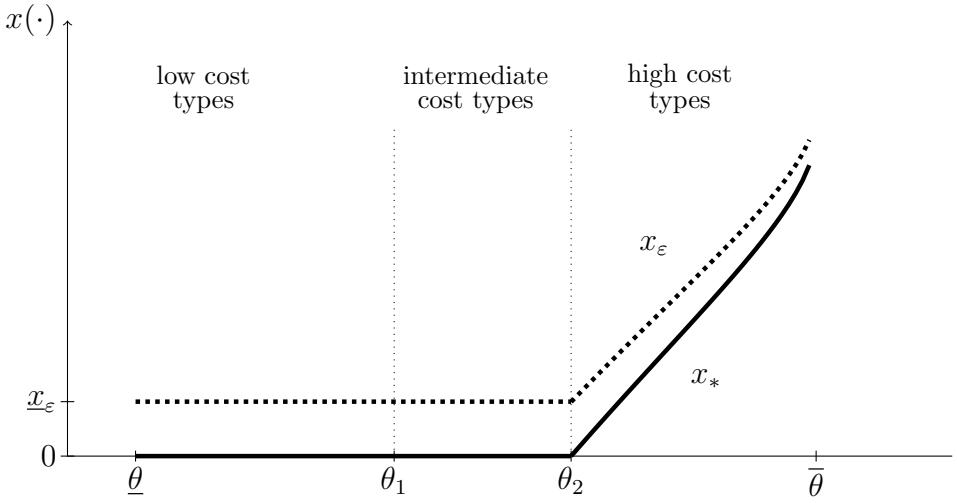


Figure 4: The inspection probability $x_*(\cdot)$ of Theorem 2.

Corollary 1. *Let $\mathbb{M}_{\bar{t}}$ be an optimal mechanism for $\mathcal{P}_0$ subject to the additional constraints $t^N(\cdot) \leq \bar{t}, t^I(\cdot) \leq \bar{t}$. Suppose the point-wise limits of $(x_{\bar{t}}(\theta), q^I_{\bar{t}}(\theta), q^N_{\bar{t}}(\theta))$ as $\bar{t} \to \infty$ exist. Then*

$$\lim_{\bar{t}\to\infty} \left(x_{\bar{t}}(\theta), q^I_{\bar{t}}(\theta), q^N_{\bar{t}}(\theta)\right) = \left(x_*(\theta), q^I_*(\theta), q^N_*(\theta)\right).$$

Theorem 2 states that the quantity allocation $(q^I_*(\cdot), q^N_*(\cdot))$ of the constrained optimal mechanism is part of an ε-optimal mechanism for any $\varepsilon > 0$. Moreover, there exists an inspection policy $x_*(\cdot)$ such that there is an ε-optimal mechanism with quantity allocation $(q^I_*(\cdot), q^N_*(\cdot))$ and an inspection probability $x_\varepsilon(\cdot)$ that is a scaled version of $x_*(\cdot)$. This scaling factor depends only on ε; in particular, it does not depend on the type θ. Taken together, this implies: one can approximate the value of the Principal's problem arbitrarily closely with a sequence of mechanisms such that the quantity allocation is constant along the sequence and the inspection probability scales.

The inspection policy $x_*(\cdot)$ is the limit of the optimal inspection policy in the constrained problem as the lower bound on the inspection policy vanishes. This inspection policy is equal to 0 for low and intermediate cost types, i.e., $\theta \leq \theta_2$ for the threshold θ_2 of Theorem 1. For high-cost types, $\theta > \theta_2$, $x_*(\cdot)$ satisfies the differential equation (6) with boundary condition $x_*(\theta_2) = 0$.

To see why the first part of Theorem 2 is true, recall that the quantity allocation in the constrained optimal mechanism does not depend on the lower bound $\underline{x}$. Since the value of the constrained problem converges to the value of the unconstrained problem, for every $\varepsilon > 0$ there exists a lower bound $\underline{x}_\varepsilon > 0$ such that the constrained optimal mechanism for this lower bound is ε-optimal in the unconstrained problem. Figure 4 shows the inspection policy $x_*(\cdot)$ and the inspection policy of the constrained optimal mechanism for $\underline{x}_\varepsilon$.

The second part of Theorem 2 states that the quantities of Theorem 1 and the inspection policy $x_*(\cdot)$ of the first part of Theorem 2 are the limit quantities and inspection probability in any ε-optimal mechanism as ε vanishes.[32] As a consequence, qualitative predictions about the quantity allocation and inspection probability do not depend on which sequence of mechanisms that approach the value W_0 we consider. In particular, we can restrict ourselves to an analytically convenient way to characterize the limit of nearly optimal mechanisms. Imposing a positive lower bound on the inspection probability, as in Section 4, is a convenient way to characterize nearly optimal mechanisms.

Theorem 2 has the following Corollary. The optimal mechanism under the restriction of an upper bound on transfers approaches the mechanism of Theorem 2 as this upper bound increases. Note that for a given lower bound $\underline{x} > 0$, all transfers in the constrained optimal mechanism for this lower bound are bounded above by a constant $\bar{t}(\underline{x})$. Since transfers after inspection diverge to infinity for some types as the lower bound $\underline{x}$ approaches 0, to each unbounded sequence of upper bounds on transfers there corresponds a vanishing sequence of lower bounds on the inspection probability. The value of the Principal's problem with the upper bound on transfers $\bar{t}(\underline{x})$ is at least the value of the constrained problem with lower bound $\underline{x}$. Consequently, as the upper bound on transfers goes to infinity, the corresponding value approaches the value of the unconstrained problem.[33]

We finish with a result about the use of quantity distortions and inspection in ε-optimal mechanisms.

Proposition 4. *1. Let W^{MR} be the Principal's payoff from the optimal mechanism when inspection is not feasible, i.e., $x(\theta) = 0$ for each type θ. W^{MR} is bounded away from the value of the unconstrained problem, irrespective of the inspection cost. Formally, there exists $\Delta > 0$ such that $W_0 - W^{MR} \geq \Delta$ for all $\kappa > 0$.*

2. For any $\kappa > 0$ there exists an $\varepsilon > 0$ such that no incentive compatible mechanism that satisfies (IR-I) and (IR-N) with $q^N(\theta) = q^{FB}(\theta)$ is ε-optimal in the unconstrained problem.

Proof in Appendix A.10.

The first part of Proposition 4 says that the Principal is strictly better off when inspection is feasible than when inspection is impossible. This is true irrespective of the magnitude of inspection costs, and the difference in values does not vanish as inspection costs grow large. The intuition behind this result is closely related to the one behind the optimal transfer schemes of Proposition 1: inspection allows the Principal to provide incentives only through payments after inspection and truthful reporting. This allows

[32] Theorem 2 does not state that the limit of the quantity allocation and inspection probability exists for every $1/n$-optimal mechanism. It does not rule out sequences of $1/n$-optimal mechanisms such that the quantity allocation or inspection probability admit no convergent subsequence.

[33] We remark that the optimal mechanism under the restriction of a lower bound on transfers $\underline{x}$ is not equal to the optimal mechanism given the corresponding upper bound on transfers $\bar{t}(\underline{x})$. In particular, quantities after inspection and a truthful report, $q^I(\theta)$, need not equal the first-best quantity $q^{FB}(\theta)$ for all types when there is an upper bound on transfers.

the Principal to pay lower transfers compared to when inspection is not feasible. Since providing incentives through rewards after truthful inspection is possible irrespective of the magnitude of the inspection probability as long as inspection probabilities are strictly positive, the Principal can reduce total transfers at negligible inspection costs, even as κ grows large. Hence, the Principal makes use of inspection in every ε-optimal mechanism.[34]

The second part of Proposition 4 says that the Principal uses quantity distortions in all ε-optimal mechanisms. Intuitively, distorting quantities without inspection downwards from first-best reduces the information rents the Principal has to pay to the Agent to induce truth-telling. While the Principal can use higher inspection probabilities instead of quantity distortions for reducing information rents, quantity distortions are less costly as long as these distortions are small enough.

6. Conclusion

This paper examines how the ability to learn the private information of a contracting party affects the optimal mechanism. In a procurement problem, a Principal can use transfers and costly inspection to induce a cost-specific allocation. We characterize the unique limit of quantities and inspection probabilities in all approximately optimal mechanisms.

Combining inspection and transfers yields new insights. First, quantity distortions for all types are not optimal. There is an interval of low cost types that produce the efficient quantity even when not inspected. Higher cost types produce less than the efficient quantity when not inspected. Both the set of types of which produce less than the efficient quantity and the size of the distortion depend on the cost of inspection. Second, inspection probabilities are positive for some types (unless the inspection cost is prohibitively large), and do not satisfy a bang-bang property. The inspection probability need not be monotone in the reported type. Moreover, restricting the mechanism to have deterministic inspection reduces the Principal's payoff. Both results are novel observations in models that combine inspection, transfers and quantities.

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[34] This does not imply that the inspection policy is positive in the limit of ε-optimal mechanisms. Indeed, this is not true; see the discussion at the end of Section 4.2 and after Lemma 4.

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A. Proofs

A.1. Proof of Proposition 1

Proof. 1. In the text.

2. Fix a mechanism $\widetilde{\mathbb{M}}$ that is incentive compatible and satisfies the constraints (IR-I) and (IR-N). Define $q^N(\cdot) = \widetilde{q^N}(\cdot)$, $q^I(\cdot,\cdot) = \widetilde{q^I}(\cdot,\cdot)$, and $x(\cdot) = \widetilde{x}(\cdot)$. Set $t^I(\hat{\theta},\theta) = q^I(\hat{\theta},\theta)\theta$ for all $\hat{\theta} \neq \theta$. For θ such that $\widetilde{x}(\theta) = x(\theta) = 0$, set $t^N(\theta) = \widetilde{t^N}(\theta)$ and $t^I(\theta,\theta) = \widetilde{t^I}(\theta,\theta)$. For θ such that $\widetilde{x}(\theta) = x(\theta) > 0$, set $t^I(\theta,\theta) = \widetilde{t^I}(\theta,\theta) + \frac{1-x(\theta)}{x(\theta)}(-\theta q^N(\theta) + \widetilde{t^N}(\theta))$ and $t^N(\theta) = \theta q^N(\theta)$. Since $\widetilde{M}$ satisfies ex-post IR constraints, i.e., $-\widetilde{q^N}(\theta)\theta + \widetilde{t^N}(\theta) \geq 0$ and $-\widetilde{q^I}(\theta,\theta)\theta + \widetilde{t^I}(\theta,\theta) \geq 0$, then $q^I(\theta,\theta)\theta - t^I(\theta,\theta) \geq 0$. Moreover, the payoff of the Principal is equal under $\mathbb{M}$ and $\widetilde{\mathbb{M}}$.

3. In the text.

□

A.2. Proof of Theorem 1

We divide the proof of the theorem into several claims.

Claim A.2.1. *For any $\underline{x} \in (0,1)$, the quantity and inspection probabilities are almost everywhere unique.*

Proof. For two functions $y : [\underline{\theta},\overline{\theta}] \to [0, 1-\underline{x}], Q : [\underline{\theta},\overline{\theta}] \to \mathbb{R}_+$, define $G(y,Q)$ to be

$$\int_{\underline{\theta}}^{\overline{\theta}} y(\theta)\Big(V\left(\frac{Q(\theta)}{y(\theta)}\right) - \theta\frac{Q(\theta)}{y(\theta)} - \Big(V(q^{FB}(\theta)) - \theta q^{FB}(\theta) - \kappa\Big)\Big) - \sup_{\hat{\theta}} Q(\hat{\theta})(\hat{\theta}-\theta)\,\mathrm{d}F(\theta).$$

The proof follows from the concavity of the map $(y,Q) \mapsto G(y,Q)$, and the fact that $q^N(\theta)$ solves (5) when $0 < y(\theta) < 1 - \underline{x}$. Details can be found in Ahmadzadeh and Waizmann (2024). □

Claim A.2.2. *All types produce the first-best quantity after inspection and truth-telling: $\forall\theta, q^I(\theta) = q^{FB}(\theta)$*

Proof. In the text □

Claim A.2.3. *Low-cost types produce the first-best quantity when not inspected: $\forall\theta \in [\underline{\theta},\theta_1], q^N(\theta) = q^{FB}(\theta)$. Moreover, low cost types are inspected with the minimal probability: $\forall\theta \in [\underline{\theta},\theta_1], x(\theta) = \underline{x}$.*

Proof. For $\delta > 0$ denote $B_\delta(\theta) = \{\hat{\theta} | (1-x(\hat{\theta}))q^N(\hat{\theta})(\hat{\theta}-\theta) \geq \pi(\theta) - \delta\}$. Define

$$\theta_1 = \inf_{\theta} \bigcap_{\delta>0} \bigcup_{\theta\in[\underline{\theta},\overline{\theta}]} B_\delta(\theta).$$

For all θ and $\hat{\theta}$, define $\chi(\hat{\theta},\theta) = (1-x(\hat{\theta}))q^N(\hat{\theta})(\hat{\theta}-\theta)$. We show that $\theta_1 > \underline{\theta}$. In any optimal mechanism, $q^N(\theta) \leq q^{FB}(\theta)$ and $\pi(\underline{\theta}) > 0$. Hence, for a $\delta > 0$ small enough there exists $\theta^\dagger > \underline{\theta}$ such that $\pi(\underline{\theta}) - \chi(\hat{\theta},\underline{\theta}) \leq 2\delta$ for all $\hat{\theta} < \theta^\dagger$. Since $\pi(\cdot)$ and $\chi(\hat{\theta},\cdot)$ are continuous in θ, $\pi(\theta) - \chi(\hat{\theta},\theta) \leq \delta$ for all $\hat{\theta} < \theta^\dagger$ and $\theta \leq \theta^{\dagger\dagger}$ or a $\theta^{\dagger\dagger} > \underline{\theta}$. Since $\chi(\hat{\theta},\theta) < 0$ for all $\theta > \hat{\theta}$, we conclude that $\theta_1 > \underline{\theta}$. Therefore given $\hat{\theta}$, there exists $\delta > 0$ such that $\pi(\theta) - \chi(\hat{\theta},\theta) > \delta$ for all θ, and $\hat{\theta} < \theta_1$. Hence there exists β° such that for all $\beta \in (\beta^\circ, -\beta^\circ)$, and for all θ

$$\pi(\theta) - (1+\beta)\chi(\hat{\theta},\theta) > \frac{\delta}{2}.$$

Now maximize point-wise: an admissible variation for $q^N(\theta)$ is $(1+\beta)q^N(\theta)$ for $\beta \in (\beta^\circ, -\beta^\circ)$. This variation does not change $\pi(\theta)$, so it changes only $V(q^N(\theta)) - \theta q^N(\theta)$ in the objective of the reduced problem. Since $V(q) - \theta q$ has a unique maximizer for all θ, we conclude $q^N(\theta) = q^{FB}(\theta)$ for all $\theta < \theta_1$. Moreover, for all $\theta < \theta_1$

$$V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa < V(q^N(\theta)) - q^N(\theta)\theta.$$

If $x(\theta) > \underline{x}$, by a similar argument using an admissible variation of $x(\theta)$ to $(1+\beta)x(\theta)$ for $\beta < 0$, we can increase the objective, without changing $\pi(\theta)$ for all $\theta \in [\underline{\theta}, \overline{\theta}]$. □

Claim A.2.4. *For all types θ such that $\underline{x} < x(\theta) < 1$ the quantity without inspection is strictly less than first-best and strictly decreasing in θ and it is given as the unique solution $q = q^N(\theta)$ to*

$$V(q^{FB}(\theta)) - \theta q^{FB}(\theta) - \kappa = V(q) - qV'(q).$$

Proof. In the text. □

Claim A.2.5. *Intermediate cost types are inspected with the minimal probability $\underline{x}$ and produce a quantity strictly less than first-best: there exists a $\theta_2, \theta_1 < \theta_2 \leq \overline{\theta}$, such that $x(\theta) = \underline{x}$ and $q^N(\theta) < q^{FB}(\theta)$ for all types $\theta \in (\theta_1, \theta_2]$.*

Proof. By Lemma 1, $Q^N(\cdot) = (1-x(\cdot))q^N(\cdot)$ is continuous, so that $\lim_{\theta \searrow \theta_1}(1-x(\theta))q^N(\theta) = (1-\underline{x})q^{FB}(\theta_1)$ from Claim A.2.3. Moreover, $q^N(\cdot) \leq q^{FB}(\cdot)$ and $x(\cdot) \geq \underline{x}$ imply that $q^N(\theta_1) = q^{FB}(\theta_1)$. Suppose toward a contradiction that for every n large enough there exists $\tilde{\theta}_n \in [\theta_1, \theta_1 + 1/n)$ with $x(\tilde{\theta}_n) > \underline{x}$. Then $q^N(\tilde{\theta}_n) = q^*(\tilde{\theta}_n)$ where q^* is the optimal quantity from Claim A.2.4. By the definition of $q^*(\cdot)$ and $q^{FB}(\cdot)$, $\inf_{\theta \in [\underline{\theta}, \overline{\theta}]} q^{FB}(\theta) - q^*(\theta) > 0$. Consequently, $\limsup_n (1-x(\tilde{\theta}_n))q^N(\tilde{\theta}_n) < (1-\underline{x})q^{FB}(\theta_1)$, in contradiction to continuity of $(1-x(\cdot))q^N(\cdot)$. Thus, there exist $\theta_2 > \theta_1$ such that $x(\theta) = \underline{x}$ for all $\theta_1 \leq \theta \leq \theta_2$. □

Claim A.2.6. *Suppose $x(\hat{\theta}) = \underline{x}$ for $\hat{\theta} \in [\theta', \theta^\dagger]$ for some $\theta' \geq \theta_1$. Then the quantity without inspection $q(\hat{\theta}) \equiv q^N(\hat{\theta})$ for types $\hat{\theta} \in [\theta', \theta^\dagger]$ solves*

$$-q'(\hat{\theta})(\hat{\theta} - \theta) = q(\hat{\theta}), \qquad \text{where } \hat{\theta} = \hat{\theta}(\theta) \text{ solves (3)},$$

for $\theta \in [\hat{\theta}^{-1}(\theta'), \hat{\theta}^{-1}(\theta^\dagger)]$ with an appropriate boundary condition for $\hat{\theta}(\cdot)$ and $q(\cdot)$.

Proof. By definition $\hat{\theta}(\theta) = \arg\max_{\tilde{\theta}\in[\underline{\theta},\overline{\theta}]} (1-x(\tilde{\theta}))q(\tilde{\theta})(\tilde{\theta}-\theta)$. By Lemma 1, $\hat{\theta}(\cdot)$ is strictly increasing. Consequently, the solution of $\max_{\tilde{\theta}\in[\underline{\theta},\overline{\theta}]} (1-x(\tilde{\theta}))q(\tilde{\theta})(\tilde{\theta}-\theta)$ cannot be a corner solution for $\theta < \overline{\theta}$. Writing the first order condition for θ such that $\hat{\theta}(\theta) \leq \theta_2$, and using the fact that $x(\cdot)$ is constant in a neighborhood of $\hat{\theta}(\theta)$, we obtain $(q)'(\hat{\theta}(\theta)) \times (\hat{\theta}(\theta)-\theta) + q(\hat{\theta}(\theta)) = 0$. □

Claim A.2.7. *In the constrained optimal mechanism, no type is inspected with probability* 1.

Proof. Let $\tilde{\theta} = \min\{\theta|\pi(\theta)=0\}$.

Step 1: if $\kappa > 0$, then full inspection, $x(\theta)=1$ for all θ, is not optimal. Set $x(\theta) = \underline{x}$ for $\theta \leq \check{\theta}$, and $x(\theta) = 1$ for $\theta > \check{\theta}$. Set $q(\theta) = q^{FB}(\theta)$ for all θ. We show that there exists $\check{\theta} > \underline{\theta}$ such that the value of the policy defined above exceeds that of full inspection. For this purpose, we should show

$$\int_{\underline{\theta}}^{\overline{\theta}} \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta\right) \mathrm{d}F(\theta) + \int_{\underline{\theta}}^{\check{\theta}} -\underline{x}\kappa \,\mathrm{d}F(\theta) - \int_{\check{\theta}}^{\overline{\theta}} \kappa \,\mathrm{d}F(\theta)$$
$$- \int_{\underline{\theta}}^{\check{\theta}} \sup_{\hat{\theta}\in[\underline{\theta},\check{\theta}]} (1-\underline{x}) q^{FB}(\hat{\theta})(\hat{\theta}-\theta)\,\mathrm{d}F(\theta) > \int_{\underline{\theta}}^{\overline{\theta}} \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta\right) \mathrm{d}F(\theta) - \kappa.$$
$$\Rightarrow (1-\underline{x}) \int_{\underline{\theta}}^{\check{\theta}} \left(\kappa - \sup_{\hat{\theta}\in[\underline{\theta},\check{\theta}]} q^{FB}(\hat{\theta})(\hat{\theta}-\theta)\right) \mathrm{d}F(\theta) > 0.$$

By letting $\check{\theta} \to \underline{\theta}$, the integrand becomes positive for some $\check{\theta} > \underline{\theta}$. Therefore there exists $\check{\theta}$ such that the above inequality holds.

Step 2: if $\kappa > 0$, then $\tilde{\theta} = \overline{\theta}$. Define $x_{[\theta',\overline{\theta}]}$, and $q_{[\theta',\overline{\theta}]}$, the solution to the problem $P_{[\theta',\overline{\theta}]}$, and define the value of this problem $W_{[\theta',\overline{\theta}]}$, where $P_{[\theta',\overline{\theta}]}$ is

$$\max_{x(\cdot)\in[\underline{x},1],q(\cdot)} \int_{\theta'}^{\overline{\theta}} x(\theta)\Big(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa\Big) + (1-x(\theta))\Big(V(q(\theta)) - q(\theta)\theta\Big)$$
$$- \sup_{\hat{\theta}} (1-x(\hat{\theta}))q(\hat{\theta})(\hat{\theta}-\theta)\,\mathrm{d}F(\theta),$$

and $\theta' \in [\underline{\theta},\overline{\theta}]$. We show if $\tilde{\theta} < \overline{\theta}$, then $W_{[\tilde{\theta},\overline{\theta}]} = \int_{\tilde{\theta}}^{\overline{\theta}} \Big(V(q^{FB}(\theta) - q^{FB}(\theta)\theta - \kappa\Big)\,\mathrm{d}F(\theta)$. The above equality means that the solution of the problem $P_{[\tilde{\theta},\overline{\theta}]}$ is full inspection. This is a contradiction to the first step.

Toward a contradiction assume $\tilde{\theta} < \overline{\theta}$. The solution of the problem $P_{[\underline{\theta},\overline{\theta}]}$ is $x_{[\underline{\theta},\overline{\theta}]}$, and $q_{[\underline{\theta},\overline{\theta}]}$ which by abuse of notation we say x, and q. Since the inspection probability is equal to 1 for types above $\tilde{\theta}$, we can say the solution of $P_{[\underline{\theta},\overline{\theta}]}$ for $\theta \geq \tilde{\theta}$ is $x(\theta) = 1$, and $\hat{q}(\theta)$, where $\hat{q}(\theta)$ can be any function (since it is irrelevant). So we can say the solution

of the problem $P_{[\underline{\theta},\overline{\theta}]}$ is $x(\theta)$, $q(\theta)$ for $\theta < \tilde{\theta}$, and for $\theta \geq \tilde{\theta}$, is $1-\beta(1-x_{[\tilde{\theta},\overline{\theta}]}(\theta))$, $q_{[\tilde{\theta},\overline{\theta}]}(\theta)$ when $\beta = 0$. Define

$$\check{x}(\theta,\beta) = \begin{cases} x(\theta) & \theta < \tilde{\theta}, \\ 1-\beta(1-x_{[\tilde{\theta},\overline{\theta}]}(\theta)) & \theta \geq \tilde{\theta}, \end{cases} \qquad \check{q}^N(\theta) = \begin{cases} q(\theta) & \theta < \tilde{\theta}, \\ q_{[\tilde{\theta},\overline{\theta}]}(\theta) & \theta \geq \tilde{\theta}. \end{cases}$$

Define $W_{[\underline{\theta},\overline{\theta}]}(\beta)$

$$\int_{\underline{\theta}}^{\overline{\theta}} \check{x}(\theta,\beta)\Big(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa\Big) + (1-\check{x}(\theta,\beta))\Big(V(\check{q}^N(\theta)) - \check{q}^N(\theta)\theta\Big) \\ - \sup_{\hat{\theta}}(1-\check{x}(\hat{\theta},\beta))\check{q}^N(\hat{\theta})(\hat{\theta}-\theta)\,\mathrm{d}F(\theta).$$

Therefore $W_{[\underline{\theta},\overline{\theta}]}(\beta = 0) = W_{[\underline{\theta},\overline{\theta}]}$. We will show $\lim_{\beta\to 0^+} \frac{W_{[\underline{\theta},\overline{\theta}]}(\beta) - W_{[\underline{\theta},\overline{\theta}]}(0)}{\beta}$, exists and we will compute it. Note that if the limit exists, by optimality of x and q we know $\lim_{\beta\to 0^+} \frac{W_{[\underline{\theta},\overline{\theta}]}(\beta) - W_{[\underline{\theta},\overline{\theta}]}(0)}{\beta} \leq 0$. Define $\chi(\theta) = \max_{\hat{\theta}\in[\tilde{\theta},\overline{\theta}]} (1-x_{[\tilde{\theta},\overline{\theta}]}(\hat{\theta}))q^N_{[\tilde{\theta},\overline{\theta}]}(\hat{\theta})(\hat{\theta}-\theta)$. Define the set $I(\beta) = \{\theta \in [\underline{\theta},\tilde{\theta}] | \pi(\theta) \leq \beta\chi(\theta)\}$. Compute $W_{[\underline{\theta},\overline{\theta}]}(\beta)$

$$\begin{aligned} W_{[\underline{\theta},\overline{\theta}]}(\beta) &= \int_{\underline{\theta}}^{\overline{\theta}} \Big(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa\Big)\,\mathrm{d}F(\theta) \\ &+ \int_{\underline{\theta}}^{\tilde{\theta}} (1-x(s))\left(V(q(\theta)) - \theta q(\theta) - \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa\right)\right)\,\mathrm{d}F(\theta) \\ &+ \int_{\tilde{\theta}}^{\overline{\theta}} \beta(1-x_{[\tilde{\theta},\overline{\theta}]}(s))\left(V(q^N_{[\tilde{\theta},\overline{\theta}]}(\theta)) - \theta q^N_{[\tilde{\theta},\overline{\theta}]}(\theta) - \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa\right)\right)\,\mathrm{d}F(\theta) \\ &- \int_{\tilde{\theta}}^{\overline{\theta}} \beta\chi(\theta)\,\mathrm{d}F(\theta) - \int_{\underline{\theta}}^{\tilde{\theta}} \max\left(\beta\chi(\theta), \pi(\theta)\right)\,\mathrm{d}F(\theta). \end{aligned}$$

Therefore $W_{[\underline{\theta},\overline{\theta}]}(\beta) - W_{[\underline{\theta},\overline{\theta}]}(0)$ is equal to

$$\begin{aligned} &\int_{\tilde{\theta}}^{\overline{\theta}} \beta(1-x_{[\tilde{\theta},\overline{\theta}]}(s))\left(V(q^N_{[\tilde{\theta},\overline{\theta}]}(\theta)) - \theta q^N_{[\tilde{\theta},\overline{\theta}]}(\theta) - \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa\right)\right)\,\mathrm{d}F(\theta) \\ &- \int_{\tilde{\theta}}^{\overline{\theta}} \beta\chi(\theta)\,\mathrm{d}F(\theta) - \int_{\underline{\theta}}^{\tilde{\theta}} \max\left(0, \beta\chi(\theta) - \pi(\theta)\right)\,\mathrm{d}F(\theta). \end{aligned}$$

First we show $\lim_{\beta\to 0^+} \int_{\underline{\theta}}^{\tilde{\theta}} \frac{\max(0,\beta\chi(\theta)-\pi(\theta))}{\beta}\,\mathrm{d}F(\theta) = 0$. The reason is

$$0 \leq \int_{\underline{\theta}}^{\tilde{\theta}} \frac{\max\left(0, \beta\chi(\theta) - \pi(\theta)\right)}{\beta}\,\mathrm{d}F(\theta) = \int_{I(\beta)} \frac{\beta\chi(\theta) - \pi(\theta)}{\beta}\,\mathrm{d}F(\theta) \leq \int_{I(\beta)} \chi(\theta)\,\mathrm{d}F(\theta).$$

Since $\cap_{\beta>0} I(\beta) = \lim_{\beta\to 0^+} I(\beta) = \tilde{\theta}$, and $\chi(\theta)$ is abounded above, then $\lim_{\beta\to 0^+} \int_{I(\beta)} \chi(\theta)\,\mathrm{d}F(\theta) = 0$. Therefore $\lim_{\beta\to 0^+} \int_{\underline{\theta}}^{\tilde{\theta}} \frac{\max(0,\beta\chi(\theta)-\pi(\theta))}{\beta}\,\mathrm{d}F(\theta) = 0$. Thus

$\lim_{\beta\to 0^+} \frac{W_{[\underline{\theta},\overline{\theta}]}(\beta)-W_{[\underline{\theta},\overline{\theta}]}(0)}{\beta}$ is equal to

$$= \int_{\tilde{\theta}}^{\overline{\theta}} (1 - x_{[\tilde{\theta},\overline{\theta}]}(s)) \left(V(q^N_{[\tilde{\theta},\overline{\theta}]}(\theta)) - \theta q^N_{[\tilde{\theta},\overline{\theta}]}(\theta) - \left(V(q^{FB}(\theta) - q^{FB}(\theta)\theta - \kappa) \right) \right) \mathrm{d}F(\theta)$$
$$- \int_{\tilde{\theta}}^{\overline{\theta}} \chi(\theta)\, \mathrm{d}F(\theta) = W_{[\tilde{\theta},\overline{\theta}]} - \int_{\tilde{\theta}}^{\overline{\theta}} \left(V(q^{FB}(\theta) - q^{FB}(\theta)\theta - \kappa \right) \mathrm{d}F(\theta) \leq 0$$

Note that by the Dominated convergence theorem, we can transfer the limit inside of the above integrals since the inside is bounded above. We know $W_{[\tilde{\theta},\overline{\theta}]} - \int_{\tilde{\theta}}^{\overline{\theta}} \left(V(q^{FB}(\theta) - q^{FB}(\theta)\theta - \kappa \right) \mathrm{d}F(\theta) \geq 0$ since full inspection is always feasible. Therefore

$$W_{[\tilde{\theta},\overline{\theta}]} = \int_{\tilde{\theta}}^{\overline{\theta}} \left(V(q^{FB}(\theta) - q^{FB}(\theta)\theta - \kappa \right) \mathrm{d}F(\theta).$$

A contradiction. □

Claim A.2.8. *A constrained optimal mechanism in which the inspection probability is weakly increasing is uniquely characterized by the threshold $\theta_1 = \hat{\theta}(\underline{\theta})$. The threshold θ_1 defines a second threshold θ_2 given by equation (4). These thresholds divide the type space into three regimes: low cost types $[\underline{\theta}, \theta_1]$, intermediate cost types $(\theta_1, \theta_2]$, and high cost types $(\theta_2, \overline{\theta}]$. The quantity without inspection, q^N, for is given as in Claim A.2.3 for $[\underline{\theta}, \theta_1]$, and Claim A.2.4 for $(\theta_2, \overline{\theta}]$, respectively. The quantity without inspection is given by $q_1(\cdot)$ for $(\theta_1, \theta_2]$. The inspection probability equals $\underline{x}$ on $[\underline{\theta}, \theta_1]$ and $(\theta_1, \theta_2]$, and is given by the solution to (6) on $(\theta_2, \overline{\theta}]$.*

Proof. The existence of the threshold $\theta_1 < \overline{\theta}$ follows from Claim A.2.3. Uniqueness of the threshold θ_1 follows from Claim A.2.1 and the characterization in Claim A.2.5.

First, we show that $\theta_1 = \hat{\theta}(\underline{\theta})$. Suppose that $\theta_1 > \hat{\theta}(\underline{\theta})$. By Claim A.2.3, $q^N(\theta') = q^{FB}(\theta')$ for $\theta' \in (\hat{\theta}(\underline{\theta}), \theta_1)$. Fix such a θ' and let θ satisfy $\hat{\theta}(\theta) = \theta'$. By Lemma 2, $\hat{\theta}(\theta) = \theta = \theta'$. This implies that $\pi(\theta) = 0$ and, since $\pi(\cdot)$ is decreasing, $\pi(\theta^\dagger) = 0$ for all $\theta^\dagger \in [\theta', \overline{\theta}]$. The latter requires $(1 - x(\theta^\dagger))q^N(\theta^\dagger) = 0$, in contradiction to Claim A.2.7. Suppose that $\theta_1 < \hat{\theta}(\underline{\theta})$ and let $\theta \in (\theta_1, \hat{\theta}(\underline{\theta}))$. By Claim A.2.5, $q^N(\theta) < q^{FB}(\theta)$. Since $\pi(\theta) > (1 - x(\theta))(\theta - \theta')$ for all θ', an infinitesimal increase in $q^N(\theta)$ does not affect information rents. However, this change raises the value of the reduced problem, a contradiction to the optimality of the mechanism. Consequently, $\hat{\theta}(\underline{\theta}) = \theta_1$.

Define $\tilde{\theta_2} = \min\{\overline{\theta}, \inf\{\theta | x(\theta) > \underline{x}\}\}$. Denote θ_2 the unique threshold defined by (4). We show that $\theta_2 = \tilde{\theta_2}$. Observe that $x(\theta) = \underline{x}$ and the quantity without inspection equals $q^N(\theta) = q_1(\theta)$ for $\theta \in (\theta_1, \tilde{\theta_2}]$, according to Claim A.2.6.

Suppose $\tilde{\theta_2} > \theta_2$. Because $q_1(\cdot), q_2(\cdot)$ are continuous, there is a positive mass of types $\theta \in (\theta_2, \tilde{\theta_2})$ with $q^N(\theta) = q_1(\theta) < q_2(\theta)$ and $x(\theta) = \underline{x}$. A variational argument similar to the one in Claim A.2.4, shows that $q^N(\theta) < q_2(\theta)$ requires $x(\theta) = 1$ in any optimal mechanism, a contradiction.

Suppose $\theta_2 > \tilde{\theta}_2$. Then $x(\theta) > \underline{x}$ for $\theta > \theta_2$ by the definition of $\tilde{\theta}_2$ and the hypothesis that the inspection probability is weakly increasing. Moreover, $x(\theta) < 1$ by Claim A.2.7. Claim A.2.4 implies that $q^N(\theta) = q_2(\theta)$ for $\theta > \tilde{\theta}_2$. Integration of (3) yields

$$\int_{\theta \in [\theta_2, \overline{\theta}]} (1 - x(\theta)) q_2(\theta) \left(V'(q_2(\theta)) - \theta \right) \mathrm{d}F(\theta) = \int_{\hat{\theta}(\theta) \in [\theta_2, \overline{\theta}]} \pi(\theta) \, \mathrm{d}F(\theta).$$

Therefore the Principal's payoff in the reduced problem equals

$$\int_{\underline{\theta}}^{\tilde{\theta}_2} (1 - \underline{x}) \left(V(q_1(\theta)) - q_1(\theta) V'(q_1(\theta)) - \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa \right) \right) \mathrm{d}F(\theta)$$
$$+ E_\theta \left[V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa \right] \equiv P_{\tilde{\theta}_2}.$$

Now consider an alternative inspection policy such that $x(\theta) = \underline{x}$ for all $\theta \leq \theta_2$. The payoff in the reduced problem under this inspection policy is

$$P_{\theta_2} \equiv \int_{\underline{\theta}}^{\theta_2} (1 - \underline{x}) \left(V(q_1(\theta)) - \theta V'(q_1(\theta)) - \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa \right) \right) \mathrm{d}F(\theta)$$
$$+ E_\theta \left[V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa \right].$$

Since $\theta_2 > \tilde{\theta}_2$ and $V(q_1(\theta)) - \theta V'(q_1(\theta)) - \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa \right) \geq 0$, $P_{\theta_2} \geq P_{\tilde{\theta}_2}$. Since since the original inspection probability is optimal, we conclude $P_{\tilde{\theta}_2} = P_{\theta_2}$, which implies $V(q_1(\theta)) - \theta V'(q_1(\theta)) - \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa \right) = 0$ almost everywhere on $(\tilde{\theta}_2, \theta_2)$. This furthermore implies that the inspection policy $x(\theta)$ remains constant at $\underline{x}$ on the interval $(\tilde{\theta}_2, \theta_2)$ in the initial inspection policy, in contradiction to the definition of $\tilde{\theta}_2$.

By definition of $\tilde{\theta}_2$ and the hypothesis that the inspection probability is increasing, Claim A.2.4, implies that $q^N(\theta)$ is given by (5) on $(\theta_2, \overline{\theta}]$. Moreover, $x(\theta) = \underline{x}$ for $\theta \in (\theta_1, \theta_2]$, and the quantity without inspection is given as in Claim A.2.6 with the boundary conditions $\hat{\theta}(\underline{\theta}) = \theta_1$ and, by continuity, $q^N(\theta_1) = q^{FB}(\theta_1)$; that is, $q^N(\theta) = q_1(\theta)$ on $(\theta_1, \theta_2]$.

By Claim A.2.3 and Claim A.2.5 the inspection probability equals $\underline{x}$ on $[\underline{\theta}, \theta_1]$ and $(\theta_1, \theta_2]$. Now we show the inspection probability is given by the solution to (6) on $(\theta_2, \overline{\theta}]$. By Claim A.4.1, $Q^N(\cdot) = (1 - x(\cdot))q^N(\cdot)$ is strictly decreasing and differentiable. Moreover, by Lemma 1, $\hat{\theta}(\cdot)$ is strictly increasing. Consequently, for every θ, $\hat{\theta}(\theta)$ satisfies the first–order condition. Recall that $\hat{\theta}(\cdot)$ is determined solely by the quantity without inspection, $q^N(\cdot)$: the inspection probability $x(\cdot)$ does not show up in equation (3). Hence, we can determine the optimal inspection probability as the solution to a differential equation, with an appropriate boundary condition. □

A.3. Proof of Proposition 2

Proof. Let $\overline{t}^N(\theta) = \frac{\pi(\theta)}{1-\underline{x}} + \underline{\theta} q^{FB}(\theta)$. By definition of θ_1, $\pi(\underline{\theta}) = (1 - \underline{x}) q^N(\theta_1)(\theta_1 - \underline{\theta}) \geq (1 - \underline{x}) q^{FB}(\theta)(\theta - \underline{\theta})$. Therefore $\overline{t}^N(\theta) \geq \theta q^{FB}(\theta)$. Moreover, $\overline{t}^N(\theta) = \frac{\pi(\theta)}{1-\underline{x}} + \underline{\theta} q^{FB}(\theta) <$

$\frac{\pi(\theta)}{1-\underline{x}} + \theta q^{FB}(\theta)$. By contradiction suppose $\frac{\pi(\underline{\theta})}{1-\underline{x}} + \underline{\theta} q^{FB}(\theta) \geq \frac{\pi(\theta)}{1-\underline{x}} + \theta q^{FB}(\theta)$. Therefore $\pi(\underline{\theta}) - \pi(\theta) \geq q^{FB}(\theta)(\theta - \underline{\theta})(1-\underline{x})$, or $\frac{\pi(\underline{\theta})-\pi(\theta)}{\underline{\theta}-\theta} \leq -q^{FB}(\theta)(1-\underline{x})$. Using the Mean Value Theorem for derivatives $\dot{\pi}(\theta') \leq -q^{FB}(\theta)(1-\underline{x})$ for some $\theta' \in [\underline{\theta}, \theta]$.[35] Using Envelope Theorem $-(1 - x(\hat{\theta}(\theta')))q^N(\hat{\theta}(\theta')) \leq -q^{FB}(\theta)(1-\underline{x})$. However, we know $1-\underline{x} > 1 - x(\hat{\theta}(\theta'))$ and $q^N(\hat{\theta}(\theta')) < q^{FB}(\hat{\theta}(\theta')) < q^{FB}(\theta_1) < q^{FB}(\theta)$. A contradiction. Moreover, $t^I(\theta) = \frac{\pi(\theta)-(1-\underline{x})t^N(\theta)+\theta q^{FB}(\theta)}{\underline{x}}$, therefore $t^I(\theta) > \theta q^I(\theta)$.

Now we show if $t^N(\theta)$ is optimal then $t^N(\theta) \in \left[\theta q^{FB}(\theta),\ \bar{t}^N(\theta)\right]$. By ex-post IR constraint $t^N(\theta) \geq \theta q^{FB}(\theta)$. By contradiction suppose $t^N(\theta) > \bar{t}^N(\theta)$. Then $\underline{\theta}$ yields a strictly higher payoff by mimicking θ. The reason is $\pi(\underline{\theta}) = (1-\underline{x})(\bar{t}^N(\theta) - \underline{\theta} q^{FB}(\theta)) < (1-\underline{x})(t^N(\theta) - \underline{\theta} q^{FB}(\theta))$. In addition, if $(t^N(\theta), t^I(\theta))$ is optimal then $\pi(\theta) = \underline{x} t^I(\theta) + (1-\underline{x}) t^N(\theta)$.

Finally we show if $t^N(\theta) \in \left[\theta q^{FB}(\theta),\ \bar{t}^N(\theta)\right]$ and $t^I(\theta) = \frac{\pi(\theta)-(1-\underline{x})t^N(\theta)+\theta q^{FB}(\theta)}{\underline{x}}$, then $(t^N(\theta), t^I(\theta))$ is an optimal transfer scheme. Since $\pi(\theta) = \underline{x} t^I(\theta) + (1-\underline{x}) t^N(\theta) - \theta q^{FB}(\theta)$, then the principal's payoff and type θ of the agent do not change. However, we still need to show that no type has an incentive to mimic another type. Let $\pi(\theta', \theta'')$ be the the payoff of θ' by mimicking $\theta'' \neq \theta'$. Therefore

$$\pi(\theta', \theta'') = (1-\underline{x})(t^N(\theta'') - \theta' q^{FB}(\theta'')) \leq (1-\underline{x})(\frac{\pi(\underline{\theta})}{1-\underline{x}} + \underline{\theta} q^{FB}(\theta'') - \theta' q^{FB}(\theta'')),$$

since $t^N(\theta) \leq \tilde{t}^N(\theta)$. By definition, $\pi(\underline{\theta}) = (1-\underline{x}) q^{FB}(\theta_1)(\theta_1 - \underline{\theta})$. Hence

$$(1-\underline{x})(\frac{\pi(\underline{\theta})}{1-\underline{x}} + \underline{\theta} q^{FB}(\theta'') - \theta' q^{FB}(\theta'')) = (1-\underline{x})(q^{FB}(\theta_1)(\theta_1 - \underline{\theta}) + q^{FB}(\theta'')(\underline{\theta} - \theta')).$$

Since $\theta'' \leq \theta_1$ and $\theta' \leq \theta_1$, then $q^{FB}(\theta_1)(\theta' - \underline{\theta}) \leq q^{FB}(\theta'')(\theta' - \underline{\theta})$. Therefore

$$\begin{aligned}&(1-\underline{x})(q^{FB}(\theta_1)(\theta_1 - \underline{\theta}) + q^{FB}(\theta'')(\underline{\theta} - \theta')) \leq (1-\underline{x}) q^{FB}(\theta_1)(\theta_1 - \theta') \\ &= (1-\underline{x})(t^N(\theta_1) - \theta' q^{FB}(\theta_1)) \leq \pi(\theta').\end{aligned}$$

The last equality is by the fact that $t^N(\theta_1) = \bar{t}^N(\theta_1) = \theta_1 q^{FB}(\theta_1)$ and the last inequality is by the fact that θ' weakly prefers to not mimic θ_1. □

Claim A.3.1. *For all $\theta \in (\underline{\theta}, \theta_1]$, $\lim_{\underline{x} \to 0} t^I(\theta) = +\infty$.*

Proof.

$$\begin{aligned}t^I(\theta) = \frac{\pi(\theta) - (1-\underline{x})\bar{t}^N(\theta) + \theta q^{FB}(\theta)}{\underline{x}} &= \frac{(1-\underline{x})(\frac{\pi(\theta)-\pi(\underline{\theta})}{1-\underline{x}}) + (\theta - \underline{\theta}) q^{FB}(\theta)}{\underline{x}} + \underline{\theta} q^{FB}(\theta) \\ &\geq \frac{\frac{\pi(\theta)-\pi(\underline{\theta})}{1-\underline{x}} + (\theta - \underline{\theta}) q^{FB}(\theta)}{\underline{x}} + \underline{\theta} q^{FB}(\theta)\end{aligned}$$

[35] Differentiability of $\pi(\cdot)$ follows from continuity of $q^N(\cdot)$ and $x(\cdot)$ that has been established in Theorem 1.

By a similar argument as the one in Proposition 2, $\frac{\pi(\theta)-\pi(\underline{\theta})}{1-\underline{x}} + (\theta - \underline{\theta})q^{FB}(\theta) > 0$ for $\theta > \underline{\theta}$. Using Lemma 3, note that $\frac{\pi(\theta)-\pi(\underline{\theta})}{1-\underline{x}}$ is independent of $\underline{x}$. Therefore

$$\lim_{\underline{x}\to 0} t^I(\theta) \geq \lim_{\underline{x}\to 0} \frac{\frac{\pi(\theta)-\pi(\underline{\theta})}{1-\underline{x}} + (\theta - \underline{\theta})q^{FB}(\theta)}{\underline{x}} + \underline{\theta} q^{FB}(\theta) = +\infty.$$

□

A.4. Proof of Lemma 1

Proof. We divide the proof of the lemma into several claims. Note that by Claim A.2.2, we know $q^I(\theta,\theta) = q^{FB}(\theta)$ for all $\theta \in [\underline{\theta}, \overline{\theta}]$. Let $y(\theta) \coloneqq 1 - x(\theta)$, $Q(\theta) \coloneqq (1-x(\theta))q^N(\theta)$, $q(\theta) \coloneqq q^N(\theta)$.

Claim A.4.1. *Suppose $(y(\cdot), q(\cdot))$ solve the problem reduced problem. Then $Q(\cdot)$ is (almost everywhere) equal to a decreasing continuous function on $\{\theta | Q(\theta) > 0\}$.*

Proof. The proof of the claim is divided into four steps.
Step 1) Suppose the set $\{\theta | \exists \theta' > \theta \wedge Q(\theta') \geq Q(\theta) > 0\}$ has positive measure. Then $(y(\cdot), q(\cdot))$ is not optimal.
Fix θ, θ' such that $\theta' > \theta \wedge Q(\theta') > Q(\theta) > 0$. By the last inequality, $y(\theta) \neq 0$. If $q(\theta) < q^{FB}(\theta)$, change $q(\theta)$ to $\min\{q^{FB}(\theta), Q(\theta')/y(\theta)\} > q(\theta)$. Then, by definition of θ, θ', this change leaves $\sup_{\hat{\theta}} Q(\hat{\theta})(\hat{\theta} - \tilde{\theta})$ unchanged for all $\tilde{\theta}$ but raises the objective point-wise. If $Q(\theta) = Q(\theta')$ one can raise $q(\theta)$ to $\min\{q^{FB}(\theta), Q(\theta')/y(\theta) + 1/n\}$ for an $n \in \mathbb{N}$ large enough without affecting $\sup_{\hat{\theta}} Q(\hat{\theta})(\hat{\theta} - \tilde{\theta})$. If $q(\theta) = q^{FB}(\theta)$, then

$$y(\theta)\left(V(q^N(\theta)) - \theta q^N(\theta) - (V(q^{FB}(\theta)) - \theta q^{FB}(\theta) - \kappa)\right) > 0.$$

Moreover, $Q(\theta') \leq (1-\underline{x})q^{FB}(\theta')$ implies $y(\theta) < 1 - \underline{x}$. Hence, one can raise $y(\theta)$ without affecting $\sup_{\hat{\theta}} Q(\hat{\theta})(\hat{\theta} - \tilde{\theta})$ for any $\tilde{\theta}$.
If such points have positive measure, the objective increases strictly.
Step 2) Q is almost everywhere equal to a decreasing (strictly decreasing when positive) function $\tilde{Q}$ such that for all θ, $\sup_{\hat{\theta}} Q(\hat{\theta})(\hat{\theta} - \theta) = \sup_{\hat{\theta}} \tilde{Q}(\hat{\theta})(\hat{\theta} - \theta)$. Denote $A = \{\theta | \forall \theta' > \theta : Q(\theta') < Q(\theta)\}$. By Step 1, A has measure 1. Define the function $\tilde{Q}(\cdot)$ by

$$\tilde{Q}(\theta) = \begin{cases} Q(\theta) & \theta \in A, \\ \sup_{\theta' > \theta, \theta' \in A} Q(\theta'). \end{cases}$$

Note that $\tilde{Q}$ is well-defined because A has measure 1. It is straightforward to see that $\tilde{Q}(\cdot)$ has the required properties.
Step 3) Suppose $Q(\cdot)$ is decreasing. Then we may assume it is left-continuous.
Recall that decreasing functions have at most countably many discontinuity points. For each $\theta \in [\underline{\theta}, \overline{\theta}]$ define a new function $\tilde{Q}(\theta)$ by $\tilde{Q}(\theta) = \limsup_{\theta' \to \theta} Q(\theta')$. Since $Q(\cdot)$ is decreasing, $\tilde{Q}(\cdot)$ is left-continuous. Moreover, the value of the objective under $Q(\cdot)$ and

$\tilde{Q}(\cdot)$ is the same, and $Q(\theta) = \tilde{Q}(\theta)$ for almost all θ.
Henceforth, assume $Q(\cdot)$ is decreasing and left-continuous.
Step 4) Suppose $Q(\cdot)$ is left-continuous, strictly decreasing when positive and has a discontinuity. Then $Q(\cdot)$ is not optimal.
Let θ^1 be a discontinuity point: $\liminf_{\theta' \to \theta^1} Q(\theta') = \lim_{\theta' \to \theta^1 +} Q(\theta') > 0$. Since there is a discontinuity at θ^1, Q is strictly decreasing and left-continuous there exists a $\varepsilon > 0$ such that for all $\delta > 0$, $Q(\theta^1) \geq Q(\theta') + \varepsilon$, for all $\theta' \in (\theta^1, \theta^1 + \delta)$. Note that for any $\theta \in [\underline{\theta}, \overline{\theta}], \hat{\theta} \in (\theta^1, \theta^1 + \delta)$, $Q(\hat{\theta})(\hat{\theta} - \theta) \leq (Q(\theta^1) - \varepsilon)(\delta + \theta^1 - \theta) \leq Q(\theta^1)(\theta^1 - \theta)$, where the last inequality holds for all $\delta > 0$ small enough. Note that, since $\pi(\theta^1) > 0$ there exists a $\delta > 0$ such that for all $\theta \in (\theta^1 - \delta, \theta^1)$, $Q(\theta^1)(\theta^1 - \theta) < \pi(\theta^1) < \pi(\theta)$. This implies that, raising $Q(\theta')$ by $\varepsilon/2$ for all $\theta' \in (\theta^1, \theta^1 + \delta)$ does not affect $\sup_{\hat{\theta}} Q(\hat{\theta})(\hat{\theta} - \theta)$ for any θ provided δ is small enough. This change, however, increases the objective, in contradiction to the optimality of Q. □
$From now we consider optimal \mathrm{x}(\cdot), q(\cdot)$ such that $Q(\cdot)$ is continuous and strictly decreasing.

Claim A.4.2. *$\hat{\theta}(\cdot)$ is upper hemicontinuous with nonempty and compact values.*

Proof. This follows from Berge's Maximum Theorem.□

Claim A.4.3. *Let $\theta' < \theta^\dagger$. Then $\sup \hat{\theta}(\theta') \leq \inf \hat{\theta}(\theta^\dagger)$.*

Proof. Assume $\check{\theta} \in \hat{\theta}(\theta')$, and $\check{\check{\theta}} \in \hat{\theta}(\theta^\dagger)$. Thus $Q(\check{\theta})(\check{\theta} - \theta') \geq Q(\check{\check{\theta}})(\check{\check{\theta}} - \theta')$, $Q(\check{\check{\theta}})(\check{\check{\theta}} - \theta^\dagger) \geq Q(\check{\theta})(\check{\theta} - \theta^\dagger)$. Therefore $(\theta^\dagger - \theta')(Q(\check{\theta}) - Q(\check{\check{\theta}})) \geq 0$. Since $Q(\cdot)$ is strictly decreasing, $\check{\theta} \leq \check{\check{\theta}}$. □

Claim A.4.4. *If $\theta' < \theta^\dagger$, and $\theta', \theta^\dagger \in \hat{\theta}(\check{\theta})$ for a type $\check{\theta}$, then $q(\theta'') = q^{FB}(\theta'')$, and $x(\theta'') = \underline{x}$ for all $\theta'' \in (\theta', \theta^\dagger)$.*

Proof. By Claim A.4.3 we know $\hat{\theta}(\theta) = [\inf \hat{\theta}(\theta), \sup \hat{\theta}(\theta)]$. The following definitions will be useful throughout the proof. For $\tilde{\gamma} > 0$, and type θ define

$$J(\tilde{\gamma}) = [\inf \hat{\theta}(\check{\theta}) + \tilde{\gamma}, \sup \hat{\theta}(\check{\theta}) - \tilde{\gamma}], \quad \chi(\theta, \tilde{\gamma}) = \sup_{\theta'' \in J(\tilde{\gamma})} \chi(\theta'', \theta),$$
$$I(\beta, \tilde{\gamma}) = \{\theta | \pi(\theta) - (1 + \beta)\chi(\theta, \tilde{\gamma}) \leq 0\}.$$

An admissible variation of $q(\theta'')$ for $\theta'' \in J(\tilde{\gamma})$ is $(1 + \beta)q(\theta'')$ for small enough $\beta > 0$. A directional derivative of the objective for $\beta > 0$ gives us

$$\lim_{\beta \to 0^+} \int_{s \in J(\tilde{\gamma})} (1 - x(s)) \left(\frac{\partial V((1 + \beta)q(s))}{\partial q(s)} q(s) - q(s)s \right) \mathrm{d}F(s)$$
$$- \lim_{\beta \to 0^+} \int_{s \in I(\beta, \tilde{\gamma})} \frac{(1 + \beta)\chi(s, \tilde{\gamma}) - \pi(s)}{\beta} \mathrm{d}F(s) \leq 0.$$

The inequality holds only if the limits above exist for each integral. We will compute the above limit, hence it exists. We claim the last integral is zero in the above inequality because

$$0 = \int_{s\in I(\beta,\tilde{\gamma})} \frac{\pi(s)-\pi(s)}{\beta}\,\mathrm{d}F(s) \leq \int_{s\in I(\beta,\tilde{\gamma})} \frac{(1+\beta)\chi(s,\tilde{\gamma})-\pi(s)}{\beta}\,\mathrm{d}F(s)$$
$$\leq \int_{s\in I(\beta,\tilde{\gamma})} \frac{(1+\beta)\pi(s)-\pi(s)}{\beta}\,\mathrm{d}F(s) = \int_{s\in I(\beta,\tilde{\gamma})} \pi(s)\,\mathrm{d}F(s).$$

If we show $\lim_{\beta\to 0^+}\int_{s\in I(\beta,\tilde{\gamma})}\pi(s)\,\mathrm{d}F(s)=0$, then by the Squeeze Theorem we conclude that $\lim_{\beta\to 0^+}\int_{s\in I(\beta,\tilde{\gamma})}\frac{(1+\beta)\chi(s,\tilde{\gamma})-\pi(s)}{\beta}\,\mathrm{d}F(s)=0$. For this purpose we show $\cap_{\beta>0}I(\beta,\tilde{\gamma}) = \lim_{\beta\to 0^+} I(\beta,\tilde{\gamma}) = \check{\theta}$. First we know $\check{\theta}\in\lim_{\beta\to 0^+} I(\beta,\tilde{\gamma})$, since $\pi(\check{\theta}) = Q(\theta'')(\theta''-\check{\theta})$ for all $\theta''\in[\inf\hat{\theta}(\theta),\sup\hat{\theta}(\theta)]$. Assume there exists $\tilde{\theta}\neq\check{\theta}$ such that $\tilde{\theta}\in\lim_{\beta\to 0^+} I(\beta,\tilde{\gamma})$. Fix $\beta>0$. This means that for all $i\in\mathbb{N}$, there exists θ_i such that $\pi(\tilde{\theta})\leq Q(\theta_i)(\theta_i-\tilde{\theta})(1+\frac{\beta}{i})$, and $\theta_i\in J(\tilde{\gamma})$. The sequence $\{\theta_i\}_{i=1}^{\infty}$ has a subsequence with a convergence point, call it $\mathring{\theta}$. Since $\frac{\beta}{i}$ converges to zero and $Q(\cdot)$ is a continuous function, we will have $\pi(\tilde{\theta})\leq Q(\mathring{\theta})(\mathring{\theta}-\tilde{\theta})$. This is a contradiction since $\mathring{\theta}\in J(\tilde{\gamma})$, and by Claim A.4.3, $J(\tilde{\gamma})\cap\hat{\theta}(\tilde{\theta})=\emptyset$. Therefore $\lim_{\beta\to 0^+} I(\beta,\tilde{\gamma}) = \check{\theta}$ for all $\tilde{\gamma}>0$. Since $\pi(\cdot)$ is bounded we can conclude $\lim_{\beta\to 0^+}\int_{s\in I(\beta,\tilde{\gamma})}\pi(s)\,\mathrm{d}F(s)=0$. Finally the directional derivative becomes

$$\lim_{\beta\to 0^+}\int_{s\in J(\tilde{\gamma})}(1-x(s))\left(\frac{\partial V((1+\beta)q(s))}{\partial q(s)}q(s)-q(s)s\right)\mathrm{d}F(s)\leq 0,$$
$$\text{or}\int_{s\in J(\tilde{\gamma})}(1-x(s))q(s)\left(\frac{\partial V(q(s))}{\partial q(s)}-s\right)\mathrm{d}F(s)\leq 0.$$

The second inequality follows from Dominated Convergence since the integrand is uniformly bounded from above. The same analysis applies to all intervals that are strictly inside $\hat{\theta}(\check{\theta})$, since $\tilde{\gamma}>0$ was arbitrary. So for all intervals $I\subset\hat{\theta}(\check{\theta})$.

$$\int_{s\in I}(1-x(s))q(s)\left(\frac{\partial V(q(s))}{\partial q(s)}-s\right)\mathrm{d}F(s)\leq 0.$$

We know $V'(q(\theta))>\theta$ for all θ such that $q(\theta)<q^{FB}(\theta)$. Therefore, except for a set of measure zero in $\hat{\theta}(\check{\theta})$, we have $q(\theta)=q^{FB}(\theta)$. By an argument similar to the one in the proof of Claim A.2.4[36], we know if $q(\theta)=q^{FB}(\theta)$, then $x(\theta)=\underline{x}$. Therefore for almost all $\theta\in\hat{\theta}(\check{\theta})$, we have $q(\theta)=q^{FB}(\theta)$, and $x(\theta)=\underline{x}$. Since $Q(\cdot)$ is a continuous function, for all $\theta\in\hat{\theta}(\check{\theta})$, we have $q(\theta)=q^{FB}(\theta)$, and $x(\theta)=\underline{x}$. □

Claim A.4.5. *Assume that there is no interval (θ',θ'') such that $q^{FB}(\theta)=1/(c_1\theta-c_2)$ for $\theta\in(\theta',\theta'')$ for positive constants $c_1>0, c_2\in[\underline{\theta},\overline{\theta}]$. Then $\hat{\theta}(\cdot)$ is single-valued on $\{\theta|\pi(\theta)>0\}$.*

[36]The argument in Claim A.2.4 does not rely on Lemma 1

Proof. If $\theta', \theta^\dagger \in \hat{\theta}(\check{\theta})$, then by Claim A.4.4, we have $\pi(\check{\theta}) = (1 - \underline{x})q^{FB}(\theta'')(\theta'' - \check{\theta})$, for all $\theta'' \in (\theta', \theta^\dagger)$. This means that $q^{FB}(\theta'') = \frac{\pi(\check{\theta})}{(1-\underline{x})(\theta''-\check{\theta})}$ which implies that $\frac{1}{q^{FB}(\theta'')} = C\theta'' + D$ where $C = \frac{1-\underline{x}}{\pi(\check{\theta})}$, and $D = \frac{-\check{\theta}(1-\underline{x})}{\pi(\check{\theta})}$ for all $\theta'' \in (\theta', \theta^\dagger)$, in contradiction to our assumption. □

Claim A.4.6. *Let $\mathbb{M}$ be an optimal mechanism such that $Q(\cdot)$ is strictly decreasing and continuous. Then $Q(\cdot)$ is differentiable for all $\hat{\theta} \in (\hat{\theta}(\underline{\theta}), \theta^\dagger)$ where $\theta^\dagger = \min\{\theta' | Q(\theta') = 0\}$.*

Proof. Fix a point $\hat{\theta} \in (\hat{\theta}(\underline{\theta}), \theta^\dagger)$. Since $Q(\cdot)$ is strictly decreasing, it is almost everywhere differentiable. Hence, there exist sequences $(\theta_i^L)_i$, and $(\theta_i^R)_i$ such that $(\hat{\theta}(\theta_i^L))_i$, and $(\hat{\theta}(\theta_i^R))_i$ converge to $\hat{\theta}$ from the left and the right, respectively, and $Q(\cdot)$ is differentiable at every point in the sequence. By definition of $\hat{\theta}(\cdot)$, necessary conditions for all for all $i \in \mathbb{N}$ are

$$Q'(\hat{\theta}(\theta_i^R))(\hat{\theta}(\theta_i^R) - \theta_i^R) + Q(\hat{\theta}(\theta_i^R)) = 0, \text{ and } Q'(\hat{\theta}(\theta_i^L))(\hat{\theta}(\theta_i^L) - \theta_i^L) + Q(\hat{\theta}(\theta_i^L)) = 0.$$

Therefore

$$Q'(\hat{\theta}(\theta_i^L)) = \frac{-Q(\hat{\theta}(\theta_i^L))}{(\hat{\theta}(\theta_i^L) - \theta_i^L)}, \text{ and } Q'(\hat{\theta}(\theta_i^R)) = \frac{-Q(\hat{\theta}(\theta_i^R))}{(\hat{\theta}(\theta_i^R) - \theta_i^R)}.$$

Since $Q(\cdot)$ and $\hat{\theta}(\cdot)$ is a continuous function, the right-hand side of both expressions converges so that $Q'(\hat{\theta}^-) = \frac{-Q(\hat{\theta})}{\hat{\theta} - \hat{\theta}^{-1}(\hat{\theta})} = Q'(\hat{\theta}^+)$. Thus $Q(\cdot)$ is differentiable at $\hat{\theta}$. □
This completes the proof of Lemma 1. □

A.5. Proof of Lemma 2

Proof. Let $q(\theta) \coloneqq q^N(\theta)$. We use three claims to prove the lemma. Define

$$\tilde{\theta} = \min\{\theta | \pi(\theta) = 0\}.$$

Let (x, q) be be the functions that attain the maximum in the reduced problem. Fix an interval $[I^-, I^+] \subset (\hat{\theta}(\underline{\theta}), \tilde{\theta})$. Let $\eta(q)(\theta) = q(\theta)\mathbb{1}_{\theta \in [I^-, I^+]}$, and $G(x, q)$ be the integrand in the objective of the reduced problem , i.e.,

$$\begin{aligned} G(x,q) =& x(\theta)\left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa\right) + (1 - x(\theta))\left(V(q(\theta)) - q(\theta)\theta\right) \\ & - \sup_{\hat{\theta}}(1 - x(\hat{\theta}))q(\hat{\theta})(\hat{\theta} - \theta). \end{aligned}$$

Define $g(\beta)$ for $\beta \in (-1, 1) \setminus \{0\}$ as $g(\beta) = \frac{G(x,q+\beta\eta) - G(x,q)}{\beta}$.

Claim A.5.1. *In an optimal mechanism,*

$$\lim_{\beta \to 0^-} g(\beta) = \int_{s \in [I^-, I^+]} (1 - x(s))q(s)\left(V'(q(s)) - s\right) \mathrm{d}F(s) - \int_{\hat{\theta}(s) \in (I^-, I^+)} \pi(s)\,\mathrm{d}F(s) \geq 0,$$

$$\lim_{\beta \to 0^+} g(\beta) = \lim_{\beta \to 0^-} g(\beta) - \int_{\hat{\theta}(s) \in \{I^-, I^+\}} \pi(s)\,\mathrm{d}F(s) \leq 0.$$

Proof. We will compute $\lim_{\beta\to 0^-} g(\beta)$, and $\lim_{\beta\to 0^+} g(\beta)$. If limits exist, then

$$\lim_{\beta\to 0^-} g(\beta) \geq 0, \quad \text{and} \quad \lim_{\beta\to 0^+} g(\beta) \leq 0.$$

First, suppose $\beta < 0$. Define

$$\chi(\theta) = \max_{\hat{\theta}\in[\underline{\theta},\overline{\theta}]\setminus(I^-,I^+)} (1 - x(\hat{\theta}))q(\hat{\theta})(\hat{\theta} - \theta).$$

Note that $\chi(\theta)$ is well defined since $[\underline{\theta},\overline{\theta}] \setminus (I^-, I^+)$ is compact and $Q(\cdot)$ is continuous. Define set $I(\beta) = \{\hat{\theta}(\theta)|\chi(\theta) \leq (1+\beta)\pi(\theta)\}$. A directional derivative for $\beta < 0$ gives us

$$\begin{aligned}\lim_{\beta\to 0^-} g(\beta) = &\int_{s\in[I^-,I^+]} (1 - x(s)) \left(V'(q(s))q(s) - q(s)s\right) \mathrm{d}F(s) \\ &- \lim_{\beta\to 0^-} \int_{\hat{\theta}(s)\in I(\beta)} \pi(s)\,\mathrm{d}F(s) - \lim_{\beta\to 0^-} \int_{\hat{\theta}(s)\in(I^-,I^+)\setminus I(\beta)} \frac{\chi(s) - \pi(s)}{\beta}\,\mathrm{d}F(s) \\ &- \lim_{\beta\to 0^-} \int_{\hat{\theta}(s)\in[\underline{\theta},\overline{\theta}]\setminus(I^-,I^+)} \frac{\pi(s) - \pi(s)}{\beta}\,\mathrm{d}F(s).\end{aligned}$$

We show the above limit for each integral exists, and we compute it. For the first integral, the limit can go inside the integral since inside is uniformly bounded above. Note that after the variation $\pi(\theta)$ for types θ such that $\hat{\theta}(\theta) \in [\underline{\theta},\overline{\theta}] \setminus (I^-, I^+)$ does not change (this corresponds to the last integral).[37] Therefore we can rewrite

$$\begin{aligned}\lim_{\beta\to 0^-} g(\beta) = &\int_{s\in[I^-,I^+]} (1 - x(s)) \left(V'(q(s))q(s) - q(s)s\right) \mathrm{d}F(s) \\ &- \lim_{\beta\to 0^-} \int_{\hat{\theta}(s)\in I(\beta)} \pi(s)\,\mathrm{d}F(s) - \lim_{\beta\to 0^-} \int_{\hat{\theta}(s)\in(I^-,I^+)\setminus I(\beta)} \frac{\chi(s) - \pi(s)}{\beta}\,\mathrm{d}F(s).\end{aligned}$$

we now show that the last integral equals zero. The reason is

$$\begin{aligned}0 = &\int_{\hat{\theta}(s)\in(I^-,I^+)\setminus I(\beta)} \frac{\pi(s) - \pi(s)}{\beta}\,\mathrm{d}F(s) \leq \int_{\hat{\theta}(s)\in(I^-,I^+)\setminus I(\beta)} \frac{\chi(s) - \pi(s)}{\beta}\,\mathrm{d}F(s) \\ \leq &\int_{\hat{\theta}(s)\in(I^-,I^+)\setminus I(\beta)} \frac{(1+\beta)\pi(s) - \pi(s)}{\beta}\,\mathrm{d}F(s) = \int_{\hat{\theta}(s)\in(I^-,I^+)\setminus I(\beta)} \pi(s)\,\mathrm{d}F(s).\end{aligned}$$

Since $\hat{\theta}(\theta)$ is a function, then $\lim_{\beta\to 0^-} I(\beta) = \cup_{\beta<0} I(\beta) = (I^-, I^+)$. By the Squeeze Theorem, we conclude

$$0 \leq \int_{\hat{\theta}(s)\in(I^-,I^+)\setminus I(\beta)} \frac{\chi(s) - \pi(s)}{\beta}\,\mathrm{d}F(s) \leq \lim_{\beta\to 0^-} \int_{\hat{\theta}(s)\in(I^-,I^+)\setminus I(\beta)} \pi(s)\,\mathrm{d}F(s) = 0.$$

Therefore

$$\lim_{\beta\to 0^-} g(\beta) = \int_{s\in[I^-,I^+]} (1 - x(s))q(s) \left(V'(q(s)) - s\right) \mathrm{d}F(s) - \int_{\hat{\theta}(s)\in(I^-,I^+)} \pi(s)\,\mathrm{d}F(s).$$

[37] Note that $\{I^-, I^+\} \subset [\underline{\theta},\overline{\theta}] \setminus (I^-, I^+)$.

For $\beta > 0$, define set $I(\beta) = \{\hat{\theta}(\theta) | (1+\beta)\chi(\theta) \geq \pi(\theta)\}$, where

$$\chi(\theta) = \max_{\tilde{\theta} \in [I^-, I^+]} (1 - x(\tilde{\theta}))q(\tilde{\theta})(\tilde{\theta} - \theta).$$

Directional derivative for $\beta > 0$ gives us

$$\begin{aligned}
&\lim_{\beta \to 0^+} g(\beta) = \int_{s \in [I^-, I^+]} (1 - x(s))q(s) \left(V'(q(s)) - s\right) \mathrm{d}F(s) \\
&- \lim_{\beta \to 0^+} \int_{\hat{\theta}(s) \in [I^-, I^+]} \frac{(1+\beta)\pi(s) - \pi(s)}{\beta} \mathrm{d}F(s) - \lim_{\beta \to 0^+} \int_{\hat{\theta}(s) \in [\underline{\theta}, \bar{\theta}] \setminus I(\beta)} \frac{\pi(s) - \pi(s)}{\beta} \mathrm{d}F(s) \\
&- \lim_{\beta \to 0^+} \int_{\hat{\theta}(s) \in I(\beta) \setminus [I^-, I^+]} \frac{(1+\beta)\chi(s) - \pi(s)}{\beta} \mathrm{d}F(s).
\end{aligned}$$

Similar to $\beta < 0$, we now show that the last integral equals zero

$$\begin{aligned}
0 = \int_{\hat{\theta}(s) \in I(\beta) \setminus [I^-, I^+]} \frac{\pi(s) - \pi(s)}{\beta} \mathrm{d}F(s) &\leq \int_{\hat{\theta}(s) \in I(\beta) \setminus [I^-, I^+]} \frac{(1+\beta)\chi(s) - \pi(s)}{\beta} \mathrm{d}F(s) \\
&\leq \int_{\hat{\theta}(s) \in I(\beta) \setminus [I^-, I^+]} \frac{(1+\beta)\pi(s) - \pi(s)}{\beta} \mathrm{d}F(s) = \int_{\hat{\theta}(s) \in I(\beta) \setminus [I^-, I^+]} \pi(s) \mathrm{d}F(s).
\end{aligned}$$

Since $\hat{\theta}(\theta)$ is a function, then $\lim_{\beta \to 0^+} I(\beta) = \cap_{\beta > 0} I(\beta) = [I^-, I^+]$. By the Squeeze Theorem, we conclude

$$- \lim_{\beta \to 0^+} \int_{\hat{\theta}(s) \in I(\beta) \setminus [I^-, I^+]} \frac{(1+\beta)\chi(s) - \pi(s)}{\beta} \mathrm{d}F(s) = 0.$$

Therefore

$$\lim_{\beta \to 0^+} g(\beta) = \int_{s \in [I^-, I^+]} (1 - x(s))q(s) \left(V^{'}(q(s)) - s\right) \mathrm{d}F(s) - \int_{\hat{\theta}(s) \in [I^-, I^+]} \pi(s) \mathrm{d}F(s).$$

Finally, we have

$$\begin{aligned}
&\lim_{\beta \to 0^-} g(\beta) = \\
&\int_{s \in [I^-, I^+]} (1 - x(s))q(s) \left(V^{'}(q(s)) - s\right) \mathrm{d}F(s) - \int_{\hat{\theta}(s) \in (I^-, I^+)} \pi(s) \mathrm{d}F(s) \geq 0, \\
&\lim_{\beta \to 0^+} g(\beta) = \lim_{\beta \to 0^-} g(\beta) - \int_{\hat{\theta}(s) \in \{I^-, I^+\}} \pi(s) \mathrm{d}F(s) \leq 0. \ \square
\end{aligned}$$

Claim A.5.2. *For all θ with $\pi(\theta) > 0$, $\hat{\theta}(\cdot)$ is a strictly increasing function.*

Proof. We will consider two steps. In step 1, we show the correspondence $\hat{\theta}^{-1}(\cdot)$ is a function (hence strictly increasing) in $(\hat{\theta}(\underline{\theta}), \tilde{\theta})$. In step 2, we show $\hat{\theta}^{-1}(\hat{\theta})$ for $\hat{\theta} = \hat{\theta}(\underline{\theta})$ is single valued.

Step 1) For contradiction, suppose that for some $\hat{\theta} \in (\hat{\theta}(\underline{\theta}), \tilde{\theta})$, the correspondence $\hat{\theta}^{-1}(\hat{\theta})$ is not single-valued. There exists a sequence $\hat{\theta} - \delta_n^- < \hat{\theta}$ converging from left to $\hat{\theta}$, and a sequence $\hat{\theta} + \delta_n^+ > \hat{\theta}$ converging from right to $\hat{\theta}$, such that $\hat{\theta}^{-1}(\hat{\theta} + \delta_n^+)$, and $\hat{\theta}^{-1}(\hat{\theta} - \delta_n^-)$ are single valued for all $n \in \mathbb{N}$.[38] Let $I_n^+ = \hat{\theta} + \delta_n^+$, and $I_n^- = \hat{\theta} - \delta_n^-$. Therefore $\int_{\hat{\theta}(s) \in \{I_n^-, I_n^+\}} \pi(s) \, \mathrm{d}F(s) = 0$. Hence $\lim_{\beta \to 0^+} g(\beta, n) = \lim_{\beta \to 0^-} g(\beta, n) = 0$, using Claim A.5.1 implies (by abusing of notation $g(\beta, n)$ is defined similar to $g(\beta)$ in Claim A.5.1, but for interval $[I_n^-, I_n^+]$)

$$\int_{\hat{\theta}(s) \in [I_n^-, I_n^+]} (1 - x(s)) q(s) \left(V'(q(s)) - s \right) \, \mathrm{d}F(s) = \int_{\hat{\theta}(s) \in [I_n^-, I_n^+]} \pi(s) \, \mathrm{d}F(s).$$

We know $\cap_{n \in \mathbb{N}} [I_n^-, I_n^+] = \hat{\theta}$. When n to infinity the left side goes to zero. therefore the right-hand side should go to zero as well, but the right side will be $\int_{\hat{\theta}(s) \in \hat{\theta}} \pi(s) \, \mathrm{d}F(s)$. Since $\pi(s) > 0$, then $\hat{\theta}^{-1}(\hat{\theta})$ is single valued. Otherwise, the above integral will be strictly positive.

Step 2) The proof is analogous, with some adaptations. Note that we assume $\kappa > 0$ (otherwise $\pi(\theta) = 0$ for all θ). Then $\hat{\theta}(\underline{\theta}) \neq \tilde{\theta}$. Fix an interval $I = (\hat{\theta}(\underline{\theta}) - \delta, \hat{\theta}(\underline{\theta}) + \delta)$, and define $g(\beta)$ similar to the previous step. First, suppose $\beta < 0$. Define

$$\chi(\theta) = \max_{\hat{\theta} \in [\underline{\theta}, \bar{\theta}] \setminus I} (1 - x(\hat{\theta})) q(\hat{\theta}) (\hat{\theta} - \theta).$$

Define set $I(\beta) \subset I$ such that $I(\beta) = \{\hat{\theta}(\theta) | \chi(\theta) \leq (1 + \beta)\pi(\theta)\}$. The directional derivative for $\beta < 0$ gives us

$$\begin{aligned} \lim_{\beta \to 0^-} g(\beta) = \int_{s \in I} (1 - x(s)) \left(V'(q(s)) - q(s) s \right) \, \mathrm{d}F(s) - \lim_{\beta \to 0^-} \int_{\hat{\theta}(s) \in I(\beta)} \pi(s) \, \mathrm{d}F(s) \\ - \lim_{\beta \to 0^-} \int_{\hat{\theta}(s) \in I \setminus I(\beta)} \frac{\chi(s) - \pi(s)}{\beta} \, \mathrm{d}F(s). \end{aligned}$$

The last integral is zero since

$$\begin{aligned} 0 = \int_{\hat{\theta}(s) \in I \setminus I(\beta)} \frac{\pi(s) - \pi(s)}{\beta} \, \mathrm{d}F(s) \leq \int_{\hat{\theta}(s) \in I \setminus I(\beta)} \frac{\chi(s) - \pi(s)}{\beta} \, \mathrm{d}F(s) \\ \leq \int_{\hat{\theta}(s) \in I \setminus I(\beta)} \frac{(1 + \beta)\pi(s) - \pi(s)}{\beta} \, \mathrm{d}F(s) = \int_{\hat{\theta}(s) \in I \setminus I(\beta)} \pi(s) \, \mathrm{d}F(s). \end{aligned}$$

$\hat{\theta}(\theta)$ is a function, then $\lim_{\beta \to 0^-} I(\beta) = \cup_{\beta < 0} I(\beta) = [\hat{\theta}(\hat{\theta}), \hat{\theta}(\hat{\theta}) + \delta)$. By the Squeeze Theorem, we conclude

$$0 \leq \int_{\hat{\theta}(s) \in I \setminus I(\beta)} \frac{\chi(s) - \pi(s)}{\beta} \, \mathrm{d}F(s) \leq \lim_{\beta \to 0} \int_{\hat{\theta}(s) \in I \setminus I(\beta)} \pi(s) \, \mathrm{d}F(s) = 0.$$

[38] In fact there at most countable points $\hat{\theta}$ that $\hat{\theta}^{-1}(\hat{\theta})$ is not single valued. Therefore the sequence exists.

Therefore

$$\lim_{\beta\to 0^-} g(\beta) = \int_{s\in I} (1-x(s))q(s)\left(V^{'}(q(s))-s\right)\,\mathrm{d}F(s) - \int_{\hat{\theta}(s)\in[\hat{\theta}(\hat{\theta}),\hat{\theta}(\hat{\theta})+\delta)} \pi(s)\,\mathrm{d}F(s).$$

For $\beta > 0$, define set $I(\beta) = \{\hat{\theta}(\theta)|(1+\beta)\chi(\theta) \geq \pi(\theta)\}$, where

$$\chi(\theta) = \max_{\hat{\theta}\in I} (1-x(\hat{\theta}))q(\hat{\theta})(\hat{\theta}-\theta).$$

The same argument leads us

$$\lim_{\beta\to 0^+} g(\beta) = \int_{s\in I} (1-x(s))q(s)\left(V^{'}(q(s))-s\right)\,\mathrm{d}F(s) - \int_{\hat{\theta}(s)\in[\hat{\theta}(\hat{\theta}),\hat{\theta}(\hat{\theta})+\delta]} \pi(s)\,\mathrm{d}F(s).$$

Letting $\delta > 0$ go to zero, the left side goes to zero, therefore the right-hand side should go to zero. This implies that $\hat{\theta}^{-1}(\underline{\theta})$ is single valued. □

Claim A.5.3. *For all $(I^-, I^+) \in \mathbb{R}^2_+$ such that $[I^-, I^+] \subset [\hat{\theta}(\underline{\theta}), \tilde{\theta}]$ we have*

$$\int_{t\in[I^-,I^+]} (1-x(s))q(s)\left(V^{'}(q(s))-s\right)\,\mathrm{d}F(s) = \int_{\hat{\theta}(s)\in[I^-,I^+]} \pi(s)\,\mathrm{d}F(s),$$

Proof. Using Claims A.5.1 and A.5.2

$$\begin{aligned}
&\lim_{\beta\to 0^+} g(\beta)\\
&= \int_{s\in[I^-,I^+]} (1-x(s))q(s)\left(V^{'}(q(s))-s\right)\,\mathrm{d}F(s) - \int_{\hat{\theta}(s)\in(I^-,I^+)} \pi(s)\,\mathrm{d}F(s) \geq 0,\\
&\lim_{\beta\to 0^+} g(\beta) = \lim_{\beta\to 0^-} g(\beta) - \int_{\hat{\theta}(s)\in\{I^-,I^+\}} \pi(s)\,\mathrm{d}F(s) \leq 0.
\end{aligned}$$

We can conclude $\lim_{\beta\to 0^+} g(\beta) = \lim_{\beta\to 0^-} g(\beta) = 0$, and

$$\int_{t\in[I^-,I^+]} (1-x(s))q(s)\left(V^{'}(q(s))-s\right)\,\mathrm{d}F(s) = \int_{\hat{\theta}(s)\in[I^-,I^+]} \pi(s)\,\mathrm{d}F(s).$$

Note that I^+ can be $\tilde{\theta}$ since the above equality holds for all I^+ close to $\tilde{\theta}$, and $\int_{\hat{\theta}(s)\in\tilde{\theta}} \pi(s)\,\mathrm{d}F(s) = 0$. In addition, I^- can be $\hat{\theta}^{-1}(\underline{\theta})$ since $\hat{\theta}^{-1}(\underline{\theta})$ is single valued. Hence the above equality holds even when $I^- = \hat{\theta}(\underline{\theta})$. □

Now we prove the lemma. By Claim A.5.2, $\hat{\theta}(\cdot)$ is strictly increasing, so both $\hat{\theta}(\cdot)$ and $\hat{\theta}^{-1}(\cdot)$ are differentiable almost everywhere. By Claim A.5.3, for all $(I^-, I^+) \in \mathbb{R}^2_+$ such that $[I^-, I^+] \subset [\hat{\theta}(\underline{\theta}), \tilde{\theta}]$

$$\begin{aligned}
&\int_{s\in[I^-,I^+]} (1-x(s))\left(V'(q(s))q(s)-q(s)s\right)\,\mathrm{d}F(s)\\
&= \int_{s\in[I^-,I^+]} (1-x(s))q(s)(s-\hat{\theta}^{-1}(s))\,\mathrm{d}F(\hat{\theta}^{-1}(s)),
\end{aligned}$$

and

$$\int_{s\in[I^-,I^+]} q(s)(1-x(s))\left((V'(q(s))-s)\,f(s)-(s-\hat{\theta}^{-1}(s))\frac{d\,\hat{\theta}^{-1}(s)}{d\,s}f(\hat{\theta}^{-1}(s))\right)\mathrm{d}s=0.$$

The above equality holds for all $[I^-,I^+]\subset[\hat{\theta}(\underline{\theta}),\overline{\theta})$. Using the Fundamental Theorem of Calculus for all s such that $\pi(s)>0$, and $s\in[\hat{\theta}(\underline{\theta}),\overline{\theta})$, almost everywhere (at all differentiable points of $\hat{\theta}^{-1}(\cdot)$) we have

$$(V'(q(s))-s)\,f(s)=(s-\hat{\theta}^{-1}(s))\frac{d\,\hat{\theta}^{-1}(s)}{d\,s}f(\hat{\theta}^{-1}(s)).$$

Fix $s^*\in(\hat{\theta}(\underline{\theta}),\overline{\theta})$. We want to show $\hat{\theta}^{-1}(\cdot)$ is differentiable at s^*. We know $\hat{\theta}^{-1}(\cdot)$ is almost everywhere differentiable, so there are two sequences s_L, and s_R converging to s^* from left and right such that $\hat{\theta}^{-1}(\cdot)$ is differentiable at each point of them. So, the above equation holds at each point s_L, and s_R. Finally, the left-hand side of the equation below is continuous in s, we conclude

$$\frac{(V'(q(s))-s)\,f(s)}{(s-\hat{\theta}^{-1}(s))f(\hat{\theta}^{-1}(s))}=\left.\frac{\mathrm{d}\hat{\theta}^{-1}(s)}{\mathrm{d}s}\right|_{s\in\{s^{*+},s^{*-}\}}.$$

This shows that $\hat{\theta}^{-1}(s)$ is differentiable at s^*. □

A.6. Proof of Proposition 3

Proof. From Lemma 1, $(1-x(\cdot))q^N(\cdot)$, is strictly decreasing and differentiable. Moreover, $\hat{\theta}(\cdot)$ is single–valued. Given θ, $\hat{\theta}(\theta)=\hat{\theta}$ solves the FOC to $\max_{\hat{\theta}}(1-x(\hat{\theta}))q(\hat{\theta})(\hat{\theta}-\theta)$. Therefore,

$$\frac{x'(\hat{\theta})}{1-x(\hat{\theta})}=q'(\hat{\theta})(\hat{\theta}-\theta)+q(\hat{\theta})\geq 0$$
$$\iff \frac{1}{\hat{\theta}-\theta}\geq-\frac{q'(\hat{\theta})}{q(\hat{\theta})}.$$

1. Since $q'<0$, $q(\hat{\theta})+q'(\hat{\theta})(\hat{\theta}-\theta)\geq q(\hat{\theta})+q'(\hat{\theta})\hat{\theta}$. The RHS of the last inequality is the derivative of $\theta\mapsto\theta q(\theta)$.
2. When $V'''\leq 0$, $-q'(\hat{\theta})/q(\hat{\theta})$ is decreasing. Moreover, $\hat{\theta}-\theta\leq\overline{\theta}-\underline{\theta}$. The claim follows.
3. $q(\theta)=\left(\frac{\alpha}{\alpha\theta^{1-\frac{1}{\alpha}}-\theta^{1-\frac{1}{\alpha}}+\left(\theta^{-1/\alpha}\right)^{1-\alpha}+\alpha\kappa-\kappa}\right)^{\frac{1}{\alpha-1}}$. Since $1/(\hat{\theta}-\theta)>1/\hat{\theta}$, the inequality in the last display is satisfied if $1/\hat{\theta}\geq-q'(\hat{\theta})/q(\hat{\theta})$. Simple algebra shows that this inequality is true for $\alpha\geq 1$.

□

A.7. Proof of Lemma 3

Proof. Using the change of variables $y(\theta) = 1 - x(\theta)$, define the problem $\overline{\mathcal{P}}_{\underline{x}}$ as

$$\max_{q^N(\cdot), 0 \le y(\cdot) \le 1-\underline{x}} \int \Big\{ y(\theta) \left(V(q^N(\theta)) - q^N(\theta)\theta \right) - \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta - \kappa \right) \\ - \sup_{\hat{\theta}} y(\theta) q^N(\hat{\theta})(\hat{\theta} - \theta) \Big\} \, \mathrm{d}F(\theta)$$

Denote its value by $\overline{W}_{\underline{x}}$. We allow $\underline{x} = 0$. Note that for any $\underline{x} > 0$, the value of the (reduced problem) is equal to $W_{FB} - \kappa + \overline{W}_{\underline{x}}$.

1. Using the second part of this Lemma for any sequence $\{\underline{x}_n\} \to 0$, with $\underline{x}_n > 0$, $\lim_{n\to\infty} W_{\underline{x}_n} = W_0$. By Lemma 4,

   $$W_{\underline{x}} = W_{FB} - \kappa + \overline{W}_{\underline{x}} = W_{FB} - \kappa + (1 - \underline{x})\overline{W}_0.$$

   Therefore $W_0 = W_{FB} - \kappa + \overline{W}_0$.

   $$\begin{aligned} & W_0 - W_{FB} \\ = & \max_{q^N(\cdot), 0 \le y(\cdot) \le 1} \int \Big\{ y(\theta) \left(V(q^N(\theta)) - q^N(\theta)\theta \right) - \left(V(q^{FB}(\theta)) - q^{FB}(\theta)\theta \right) \\ & - \sup_{\hat{\theta}} y(\theta) q^N(\hat{\theta})(\hat{\theta} - \theta) - (1 - y(\theta))\kappa \Big\} \, \mathrm{d}F(\theta). \end{aligned}$$

   Each part of the above expression is weakly negative. Clearly there is no $y(\cdot)$ and q^N that makes the above expression exactly zero. Therefore $W_0 < W_{FB}$.

   To show that the unconstrained problem admits no solution, assume toward a contradiction that an optimal mechanism does exist. Clearly, an optimal mechanism is also ε-optimal so that the optimal mechanism is given as in the second part of Theorem 2. For types $\theta \in (\theta_1, \theta_2)$, $x(\theta) = 0$ and $q^N(\theta) < q^{FB}(\theta)$. Incentive compatibility requires that $t^N(\theta) > \theta q^N(\theta)$ for these types. However, this cannot be an optimal mechanism: choose inspection probability $\tilde{x(\theta)} = \delta_1 > 0$, $q^N(\theta) < \tilde{q^N}(\theta) = q^N(\theta) + \delta_2 < q^{FB}(\theta)$ such $t^N(\theta) \ge \theta(q^N(\theta) + \delta_2)$, along with setting $\tilde{t^I}(\theta)$ such that

   $$\delta_1(\tilde{t^I}(\theta) - \theta q^{FB}(\theta)) + (1 - \delta_1)(t^N(\theta) - \theta(q^N(\theta) + \delta_2) = t^N(\theta) - \theta q^N(\theta)$$

   leaves incentives for type θ unchanged. Since $t^N(\theta)$ does not change, incentives for all other types $\theta' \neq \theta$ are preserved. Letting $\delta_1 \to 0$, this strictly increases the Principal's payoff, in contradiction to the optimality of the original mechanism.

2. By continuity of $W_{\underline{x}}$ in $\underline{x}$, for any sequence $\{\underline{x}_n\} \to 0$, with $\underline{x}_n > 0$,

   $$\lim_{n\to\infty} W_{\underline{x}_n} = W_0.$$

□

A.8. Proof of Lemma 4

Proof. Using the notation $y(\theta) = (1 - x(\theta))(1 - \underline{x})$, we see that the reduced problem is equivalent to

$$\max_{y(\cdot),q^N(\cdot)} \int_{\underline{\theta}}^{\overline{\theta}} \frac{y(\theta)}{1-\underline{x}} \Big(V(q^N(\theta)) - \theta q^N(\theta) - \Big(V(q^{FB}(\theta)) - \theta q^{FB}(\theta) - \kappa \Big) \Big) - \sup_{\hat{\theta}} \frac{y(\hat{\theta})}{1-\underline{x}} q^N(\hat{\theta})(\hat{\theta} - \theta)\, \mathrm{d}F(\theta),$$

subject to $0 \le y(\hat{\theta}) \le 1$. The claim follows immediately. □

A.9. Proof of Theorem 2

Proof. Note that proof of Lemma 4 does not depend on Theorem 2. We define two auxiliary problems. Denote the following problem by $\mathcal{P}_1$, and its value by W_1.

$$\max_{\substack{x(\cdot),q^N(\cdot) \\ q^I(\cdot,\cdot),t^N(\cdot)}} \int_{\underline{\theta}}^{\overline{\theta}} x(\theta) \Big(V(q^I(\theta,\theta)) - \theta q^I(\theta,\theta) - \kappa \Big) + (1 - x(\theta)) \Big(V(q^N(\theta)) - \theta q^N(\theta) \Big) - \sup_{\hat{\theta}} (1 - x(\hat{\theta}))(t^N(\hat{\theta}) - q^N(\hat{\theta})\theta)\, \mathrm{d}F(\theta).$$

subject to $0 \le x(\theta) \le 1$ and $-q^N(\theta)\theta + t^N(\theta) \ge 0$. Denote the following problem by $\mathcal{P}_2$, and its value by W_2.

$$\max_{\substack{x(\cdot),q^N(\cdot) \\ q^I(\cdot,\cdot)}} \int_{\underline{\theta}}^{\overline{\theta}} x(\theta) \Big(V(q^I(\theta,\theta)) - \theta q^I(\theta,\theta) - \kappa \Big) + (1 - x(\theta)) \Big(V(q^N(\theta)) - \theta q^N(\theta) \Big) - \sup_{\hat{\theta}} (1 - x(\hat{\theta})) q^N(\hat{\theta})(\hat{\theta} - \theta)\, \mathrm{d}F(\theta).$$

subject to $0 \le x(\theta) \le 1$.

1. By continuity of $W_{\underline{x}}$ in $\underline{x}$, $\lim_{\underline{x}\to 0} W_{\underline{x}} = W_0$. Lemma 4 implies quantities in a constrained optimal mechanism (for $\mathcal{P}_{\underline{x}}$) do not depend on the minimal inspection probability. In addition, by Lemma 4, $x_{\underline{x}}$ and $x_{\underline{x}'}$ are related by

$$\frac{1 - x_{\underline{x}}(\theta)}{1 - \underline{x}} = \frac{1 - x_{\underline{x}'}(\theta)}{1 - \underline{x}'}.$$

   Define $1 - x_*(\theta)$ to be the above ratio. Let $t^N_\epsilon(\theta) = \theta q^N_*(\theta)$ and t^I_ϵ solves equation 1. Therefore, for every ε, we can find $\underline{x}_\varepsilon$, such that $\mathbb{M}_{\underline{x}_\varepsilon}$ satisfies all conditions of Theorem 2.

2. Note that $W_0 = W_1 = W_2$. A similar argument as the proof of Lemma 4, implies the value of problem $\mathcal{P}_{\underline{x}}$ converges to W_2 when $\underline{x}$ goes to zero. Denote $\mathbb{M}_0 = \big(x_0(\cdot), q_0^I(\cdot), t_0^I(\cdot), q_0^N(\cdot), t_0^N(\cdot)\big)$, a feasible mechanism in $\mathcal{P}_0$. Define $\mathbb{M}_1 = \big(x_0(\cdot), q_0^I(\cdot), q_0^N(\cdot), t_0^N(\cdot)\big)$, and $\mathbb{M}_2 = \big(x_0(\cdot), q_0^I(\cdot), q_0^N(\cdot)\big)$. Denote the payoff of the objective in $\mathcal{P}_i$ under mechanism $\mathbb{M}$ by $W_i(\mathbb{M})$. We proceed the proof by showing a claim.

   Claim A.9.1. *If $\mathbb{M}_0$ is a feasible mechanism in $\mathcal{P}_0$, then $\mathbb{M}_1$ is a feasible mechanism in $\mathcal{P}_1$, and $\mathbb{M}_2$ is a feasible mechanism in $\mathcal{P}_2$. Moreover, $W_0(\mathbb{M}_0) \leq W_1(\mathbb{M}_1) \leq W_2(\mathbb{M}_2)$.*

   Proof. Define $\tilde{\mathbb{M}}_0 = \big(x_0(\cdot), q_0^I(\cdot), \tilde{t}_0^I(\cdot), q_0^N(\cdot), t_0^N(\cdot)\big)$ such that for all θ, $\tilde{t}_0^I(\theta) = t_0^I(\theta)$, and for all $\hat{\theta} \neq \theta$, $\tilde{t}_0^I(\hat{\theta}, \theta) = \theta q_0^I(\hat{\theta}, \theta)$. By replacing the right side of the incentive constraint in the objective, the value of $\mathcal{P}_0$ weakly increases. This implies that $\mathbb{M}_1$ is a feasible mechanism in $\mathcal{P}_1$ and $W_0(\mathbb{M}_0) \leq W_1(\mathbb{M}_1)$. □

   Define $\tilde{\mathbb{M}}_1 = \big(x_1(\cdot), q_1^I(\cdot), q_1^N(\cdot), \tilde{t}_1^N(\cdot)\big)$, such that for all θ, $\tilde{t}_1^N(\theta) = \theta q_1^N(\theta)$. An immediate observation is $W_1(\tilde{\mathbb{M}}_1) \geq W_1(\mathbb{M}_1)$. This implies that $\mathbb{M}_2$ is a feasible mechanism in $\mathcal{P}_2$ and $W_1(\mathbb{M}_1) \leq W_2(\mathbb{M}_2)$. □.
   $An immediate observation of the above claim is as follows : suppose \mathbb{M}_{0,n}$ is a sequence of feasible and $1/n$-optimal mechanisms to problem $\mathcal{P}_0$. Then $\mathbb{M}_{1,n}$, and $\mathbb{M}_{2,n}$ are sequences of feasible and $1/n-$optimal mechanisms to problem $\mathcal{P}_1$, and $\mathcal{P}_2$ respectively. Moreover, by Lebesgue's dominated convergence theorem, the point-wise limit (if exists) of sequences $x_{0,n}(\theta), q_{0,n}^I(\theta)$, and $q_{0,n}^N(\theta)$ is the unique solution of $\mathcal{P}_2$. The proof of the uniqueness of the solution of $\mathcal{P}_2$ is by Claim A.2.1.

   □

A.10. Proof of Proposition 3

Proof. From Lemma 1, $(1-x(\cdot))q^N(\cdot)$, is strictly decreasing and differentiable. Moreover, $\hat{\theta}(\cdot)$ is single–valued. Given θ, $\hat{\theta}(\theta) = \hat{\theta}$ solves the FOC to $\max_{\hat{\theta}}(1 - x(\hat{\theta}))q(\hat{\theta})(\hat{\theta} - \theta)$. Therefore,

$$\frac{x^{'}(\hat{\theta})}{1 - x(\hat{\theta})} = q^{'}(\hat{\theta})(\hat{\theta} - \theta) + q(\hat{\theta}) \geq 0$$

$$\iff \frac{1}{\hat{\theta} - \theta} \geq -\frac{q^{'}(\hat{\theta})}{q(\hat{\theta})}.$$

1. Since $q^{'} < 0$, $q(\hat{\theta}) + q^{'}(\hat{\theta})(\hat{\theta} - \theta) \geq q(\hat{\theta}) + q^{'}(\hat{\theta})\hat{\theta}$. The RHS of the last inequality is the derivative of $\theta \mapsto \theta q(\theta)$.

2. When $V^{'''} \leq 0$, $-q^{'}(\hat{\theta})/q(\hat{\theta})$ is decreasing. Moreover, $\hat{\theta} - \theta \leq \overline{\theta} - \underline{\theta}$. The claim follows.

3. $q(\theta) = \left(\frac{\alpha}{\alpha\theta^{1-\frac{1}{\alpha}} - \theta^{1-\frac{1}{\alpha}} + \left(\theta^{-1/\alpha}\right)^{1-\alpha} + \alpha\kappa - \kappa} \right)^{\frac{1}{\alpha-1}}$. Since $1/(\hat{\theta} - \theta) > 1/\hat{\theta}$, the inequality in the last display is satisfied if $1/\hat{\theta} \geq -q'(\hat{\theta})/q(\hat{\theta})$. Simple algebra shows that this inequality is true for $\alpha \geq 1$.

□

Proof. 1. Let $(q^{MR}(\cdot), t^{MR}(\cdot))$ be the optimal mechanism when inspection is not feasible. Assumption 1 guarantees that $q^{MR}(\theta) > 0$ for all types θ.[39] By revenue equivalence, transfers satisfy $t^{MR}(\theta) = \theta q^{MR}(\theta) + \int_{\theta}^{\overline{\theta}} q^{MR}(s)\,\mathrm{d}s$. Note that incentive compatibility requires that $t^{MR}(\theta) - \theta q^{MR}(\theta) = \sup_{\hat{\theta}} t^{MR}(\hat{\theta}) - \theta q^{MR}(\hat{\theta})$.

Construct a feasible mechanism for $\mathcal{P}_0$ as follows. Set $x(\theta) = x > 0$, $q^I(\theta) = q^N(\theta) = q^{MR}(\theta)$, $t^N(\theta) = \theta q^N(\theta)$ and $t^I(\theta)$ given as in equation (1). By the same argument as in the proof of Proposition 1, this mechanism is incentive compatible and satisfies the (IR-I) and (IR-N) constraints. Note that $x(t^I(\theta) - \theta q^I(\theta)) + (1-x)(t^N(\theta) - \theta q^N(\theta)) = (1-x)\sup_{\hat{\theta}} q^{MR}(\hat{\theta})(\hat{\theta} - \theta) < (1-x)\sup_{\hat{\theta}} t^{MR}(\hat{\theta}) - \theta q^{MR}(\hat{\theta})$ since $t^{MR}(\hat{\theta}) > \hat{\theta} q^{MR}(\hat{\theta})$ for all $\hat{\theta} \neq \overline{\theta}$. As a consequence,

$$\begin{aligned} W_0 - W^{MR} &\geq \sup_{\hat{\theta}}(t^{MR}(\hat{\theta}) - \theta q^{MR}(\hat{\theta})) - (1-x)\sup_{\hat{\theta}} q^{MR}(\hat{\theta})(\hat{\theta} - \theta) - x\kappa \\ &\to_{x\to 0} \sup_{\hat{\theta}}(t^{MR}(\hat{\theta}) - \theta q^{MR}(\hat{\theta})) - \sup_{\hat{\theta}} q^{MR}(\hat{\theta})(\hat{\theta} - \theta) > 0 \end{aligned}$$

2. Let $x(\cdot)$ be the solution to

$$\max_{0\leq x(\theta)\leq 1} \int V(q^{FB}(\theta)) - \theta q^{FB}(\theta) - x(\theta)\kappa - \sup_{\hat{\theta}}(1 - x(\hat{\theta}))q^{FB}(\hat{\theta})(\hat{\theta} - \theta)\,\mathrm{d}F(\theta).$$

Clearly, $x(\theta) \neq 1$ for a positive mass of types θ. As in the proof of the second part Theorem 2, this is the maximal payoff any mechanism with $q^N(\theta) = q^{FB}(\theta)$ for all θ can attain. Consider an alternative mechanism with $q^N(\theta) = \beta q^{FB}(\theta)$ for a $\beta \leq 1$ and same inspection probabilities. Note that the Principal can get a payoff (in the limit) from such a mechanism equal to

$$\begin{aligned} \int \Big\{ & x(\theta)\left(V(q^{FB}(\theta)) - \theta q^{FB}(\theta) - x(\theta)\kappa\right) + (1 - x(\theta))\left(V(\beta q^{FB}(\theta)) - \theta\beta q^{FB}(\theta)\right) \\ & - \sup_{\hat{\theta}}(1 - x(\hat{\theta}))\beta q^{FB}(\hat{\theta})(\hat{\theta} - \theta) \Big\}\,\mathrm{d}F(\theta) \end{aligned}$$

Taking the derivative with respect to β and evaluating at $\beta = 1$ yields

$$\int -\sup_{\hat{\theta}}(1 - x(\hat{\theta}))q^{FB}(\hat{\theta})(\hat{\theta} - \theta)\,\mathrm{d}F(\theta) < 0.$$

Hence, for some $\varepsilon > 0$ there is no ε-optimal mechanism with $q^N(\theta) = q^{FB}(\theta)$ for all θ.

□

[39] The proof goes through as long as $q^{MR}(\theta) > 0$ for a positive mass of types.